ACID-BASE BALANCE

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1. Introduction

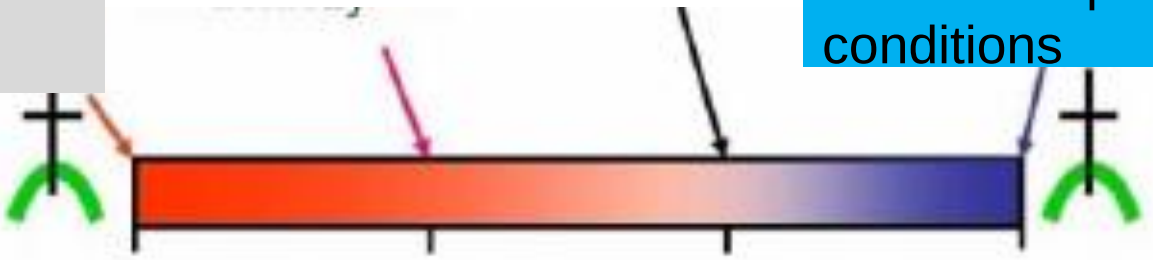
Definition of acidemia and alkalemia

Extreme value that is compatible with life under exceptional conditions

Serious acidosis

Normal pH

Extreme value that is compatible with life under exceptional conditions



pH:	6,8	7,1	7,4	7,7
pH increase		− 0,3	− 0,3	+ 0,3
antialgorithm		2x	2x	1/2x
H^+ increase		400 %	200 %	50 %

In comparison with alkalosis, acidosis can be compensated better, because we accept larger drop of pH than its increase.

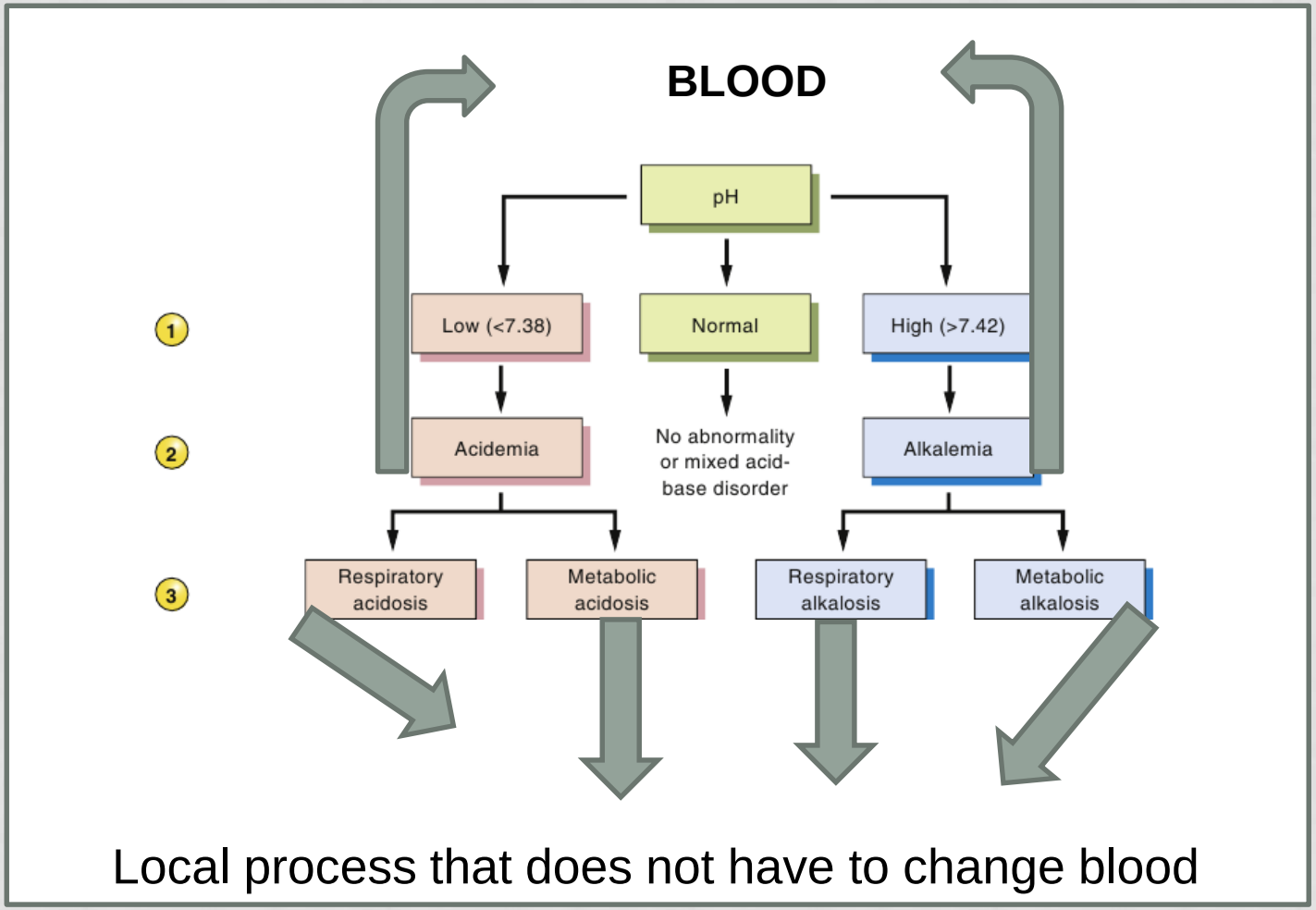
Acidemia: general effect

- Negative inotropic (contractility), and chronotropic (heart rate) effect but it is neutralized by catecholamines that are also stimulated by acidosis.
- Inhibit CNS including respiratory centers
- In systemic vessels,
  - acidosis causes vasodilation in peripheral arterioles but is also offset by catecholamine, so that peripheral vascular resistance remains relatively constant.
    - In brain, vasodilation can cause edema.
  - in the venous system, the direct effect of acidosis is vasoconstriction, which is further enhanced by catecholamine release: prevent lung edema
- Osteoporosis : Ca^{2+} and phosphate liberated from bone (hydroxyapatite, Ca5(PO4)3(OH))
- Final stage: coma (due to H^+-mediated protein denaturation in CNS, dehydration, hyperosmolarity in diabetic ketoacidosis)

Alkalemia: general effect

- Acute alkalosis increases binding and reduces free plasma Ca^{2+} leading to tetany
- Compensation of MA by hypoventilation is limited by O_2
- In brain, vasoconstriction that decrease its perfusion
- Final stage: coma or asphyxia

acidosis vs acidemia and alkalosis vs alkalemia



Blood sampling to determine the acid-base status

- Sampling:
  - arterial blood
  - capillary blood
  - venous blood
  - arterial and venous sampling

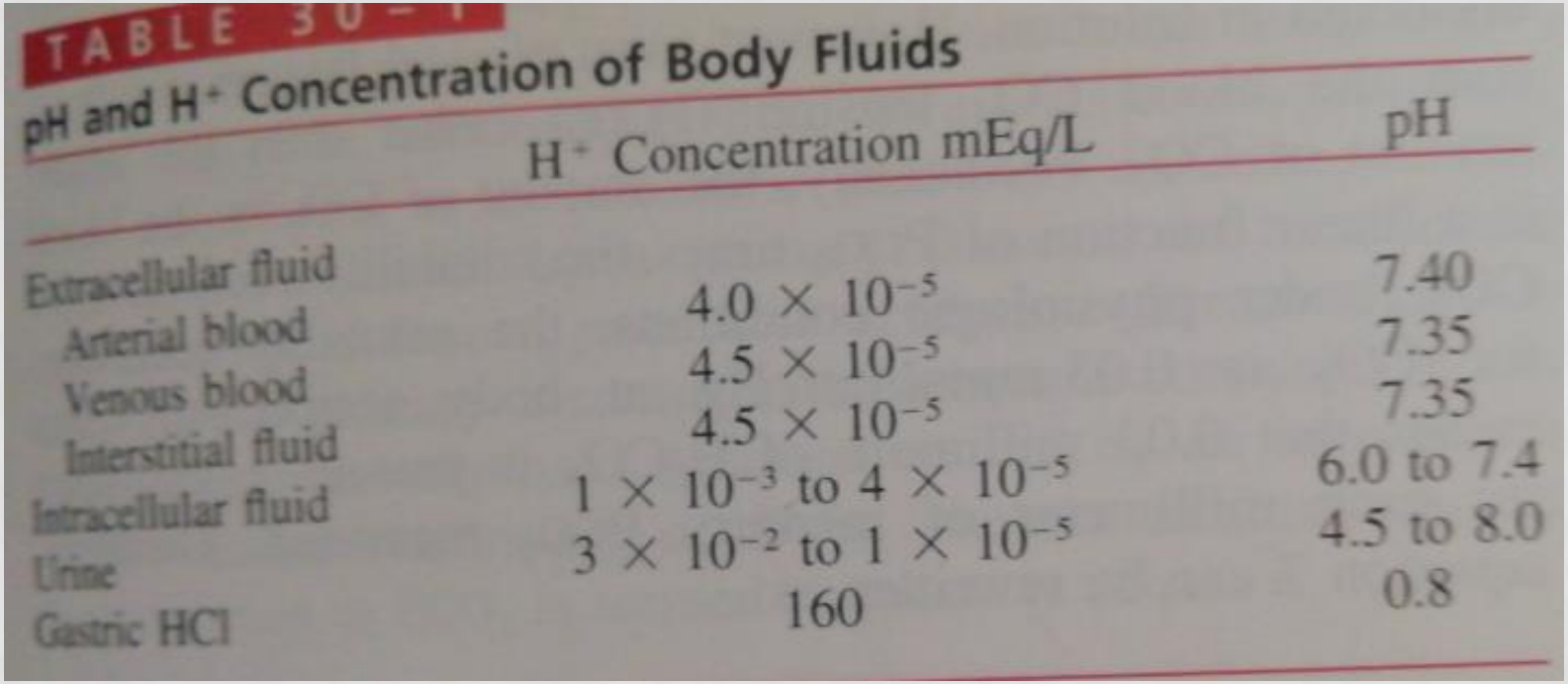


TABLE 30-1

pH and H^+ Concentration of Body Fluids

	H^+ Concentration mEq/L	pH
Extracellular fluid		
Arterial blood	4.0×10^{-5}	7.40
Venous blood	4.5×10^{-5}	7.35
Interstitial fluid	4.5×10^{-5}	7.35
Intracellular fluid	1×10^{-3} to 4×10^{-5}	6.0 to 7.4
Urine	3×10^{-2} to 1×10^{-5}	4.5 to 8.0
Gastric HCl	160	0.8

Blood sampling to determine the acid-base status

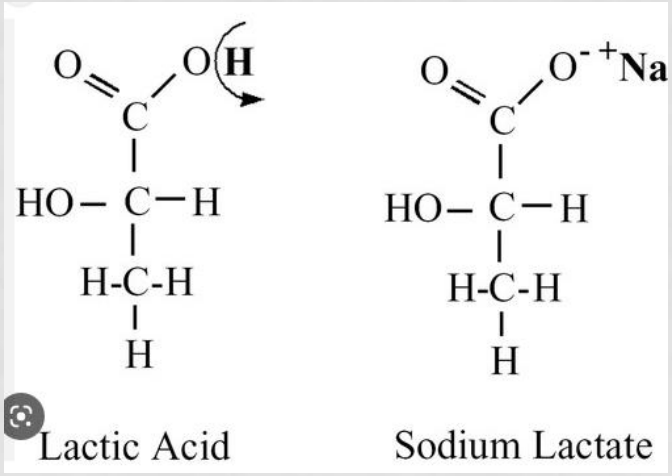
- Into heparinized capillaries immediately hermetically sealed to prevent uncontrolled contact with air (02 CO2)
- Check whether the column of blood in the capillaries is without bubbles
- The sample immediately into crushed ice or in a refrigerator (temperature up to 0^0 C) in order to immediate arrest metabolism of erythrocyte-lactate production – it may affect the evaluation of the sample ABR !!!!)
- the fastest sample processing - barrier of inadequate results

2. Examples of biochemical pathways that produce H^+

- Anaerobic glycolysis (lactic acid)
- Lipolysis
- Ketone bodies in the liver
- Degradation of some amino acids: Sulphur containing amino acids and cationic amino acids produce H_2SO_4 and H_3PO_4

Production of H+ anaerobic metabolism

- The electron transport chain cannot work without O_2, leading to accumulation of NADH. Therefore, lactic acid is synthetized to replenish NAD^+. The lactic acid create rapidly lactate and H^+.
- Lactate can be used as a source of energy.
  - Lactate undergoes oxidation (O_2 is required) predominantly in the hepatocytes, but also in the kidneys as well as cardiac and skeletal
  - Lactate undergoes gluconeogenesis, predominantly in the liver and to a lesser extent in the kidneys.



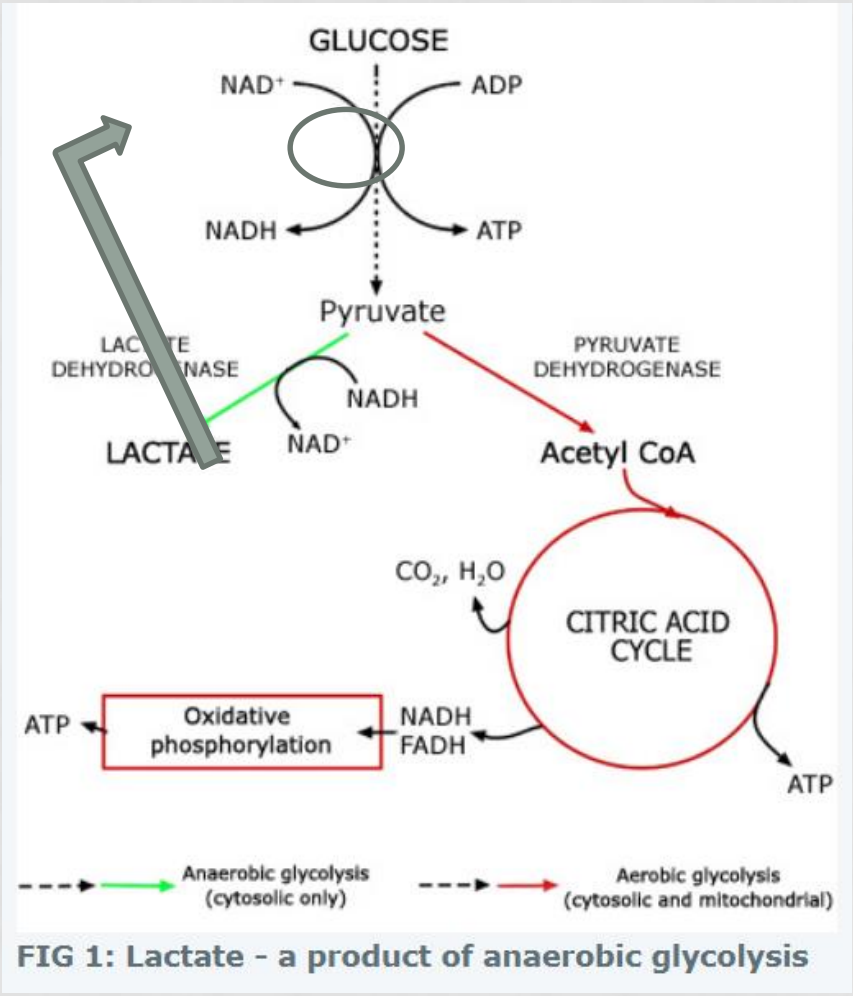


FIG 1: Lactate - a product of anaerobic glycolysis

Other sources of H+ production

- Production of free fatty acids from adipose tissue (lipolysis)
  - Triglycerides → glycerol + 3 fatty acids + $3H^+$

In the mitochondria, beta-oxidation generates one acetyl CoA, one NADH and one FADH2 for every removal of a two-carbon group from fatty acids. Acetyl-CaA enters to the citric acid cycle while NADH and FADH2 are used in the electron transport chain where O_2 is required.

Production of ketone bodies

- Ketone bodies are produced mainly in the mitochondria of liver cells in response to an unavailable of blood glucose, such as during fasting.
  - Fatty acids → ketone bodies + nH^+
- Ketone bodies are alternative source of energy, e.g. for heart
- The most important ketone bodies are acetone, acetoacetate and beta-hydroxybutyrate

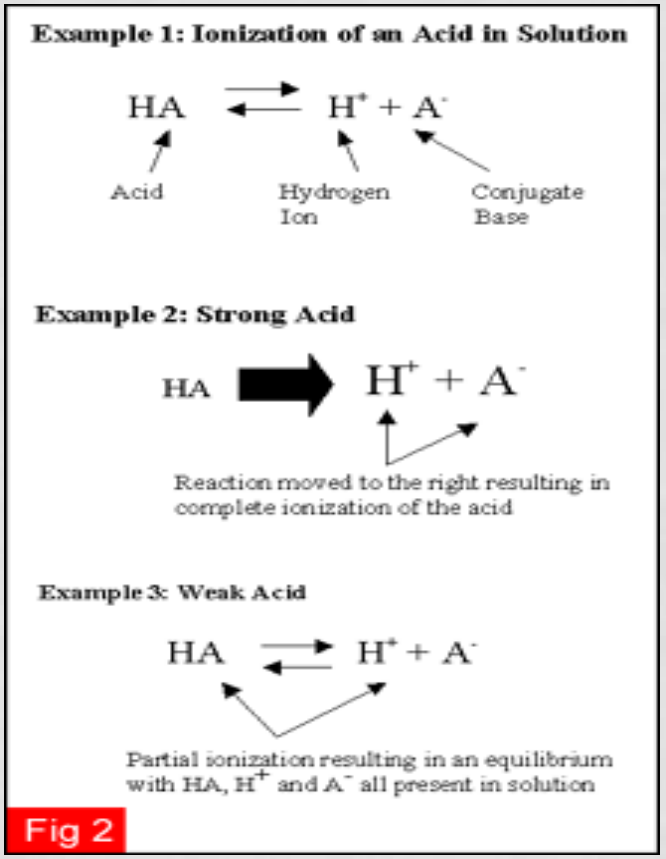
Other Sources of H+

- **H_2SO_4** - sulphuric acid is produced by oxidation of sulfhydryl groups, e.g. The degraded amino acids that contain sulphur such as cysteine, methionine.
- **H_3PO_4** - phosphoric acid is produced by hydrolysis of phosphoproteins, phospholipids, nucleic acids

3. Buffers

What is buffer: pH basics

- Acids create hydrogen ions in the solution ("proton donors")
- The bases bind hydrogen ions in solution with ("proton acceptors")
- **Strong acids** ionize completely in solution to form H^+ and base
- **Weak acids and bases** are only partially ionized in solutions
- **Buffers** are substances that reduce the changes in concentration of H^+ (pH), where H^+ added to the solution or removed therefrom: It is **a weak acid and a base.**



Buffer system and organs

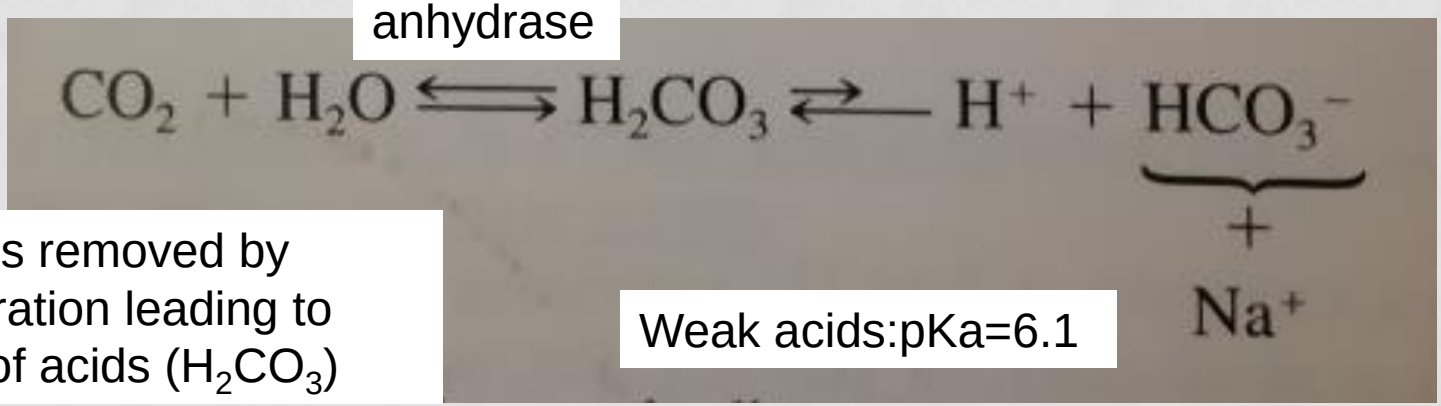
- Blood:
  - Hydrogen carbonate
  - Hemoglobin
  - Plasma protein
- Renal buffer:
  - Hydrogen phosphate
  - NH_3/NH_4^+ buffer
- Ca^{2+} bone buffer
- Liver
  - Urea vs. glutamine
- H^+/K^+ or Cl^-/HCO_3^- exchange between extra and intracellular fluid

- The bicarbonate buffer accounts for 53% of all buffers. The second most important buffer is hemoglobin (Hemoglobin is an intracellular buffer)
  - extracellular HCO_3^- (53%)
  - intracellular (18% in erythrocytes)

The bicarbonate buffer: HCO_3^-

- The bicarbonate buffer is a good buffer because of two:
  - H_2CO_3 is weak acid
  - A loss of CO_2 in the lung

Carbonic anhydrase



CO_2 is removed by respiration leading to loss of acids (H_2CO_3)

Weak acids:pKa=6.1

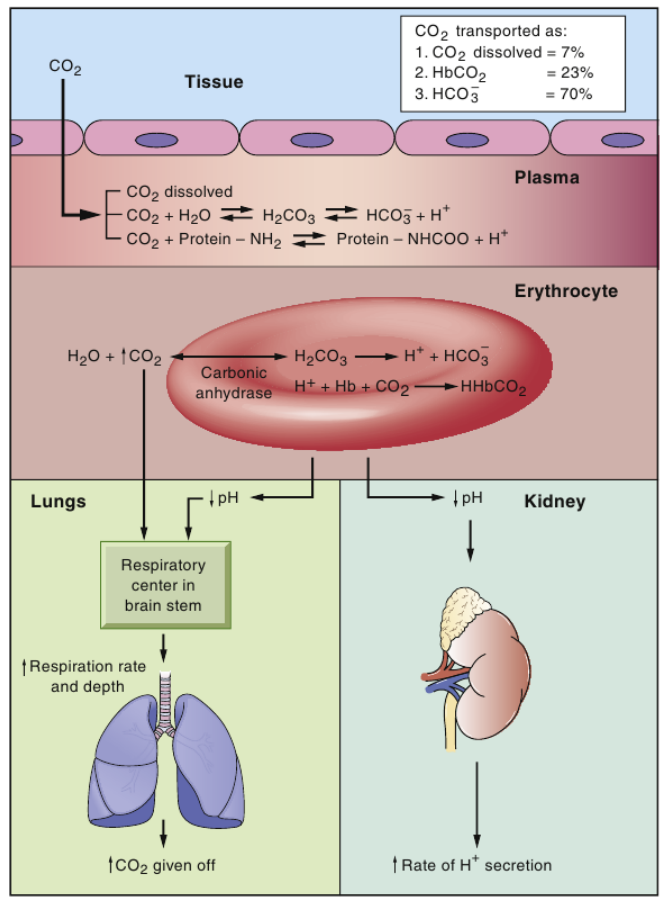
This reaction is slow unless the enzyme carbonic anhydrase is present. This enzyme is abundant in the wall of the lung alveoli, where CO_2 is released and in renal tubules, where CO_2 with H_2O reacts and form H_2CO_3.

The hemoglobin buffer is intracellular buffer

The hemoglobin buffer is main non-bicarbonate buffer (less plasmatic proteins and phosphates)

In tissue, hemoglobin liberates O_2 and binds H^+ that is formed by enzymatic (carbonic anhydrase) reaction $CO_2+H_2O = H^+ + HCO_3^-$
(HCO_3^- is release from the red cells and Cl^- enter red cells to keep neutral el. charge, venous blood has less Cl^-)

- Carbaminohemoglobin is a weaker acid (pK_A=7.8): Tissue H^+ are buffered by hemoglobin.



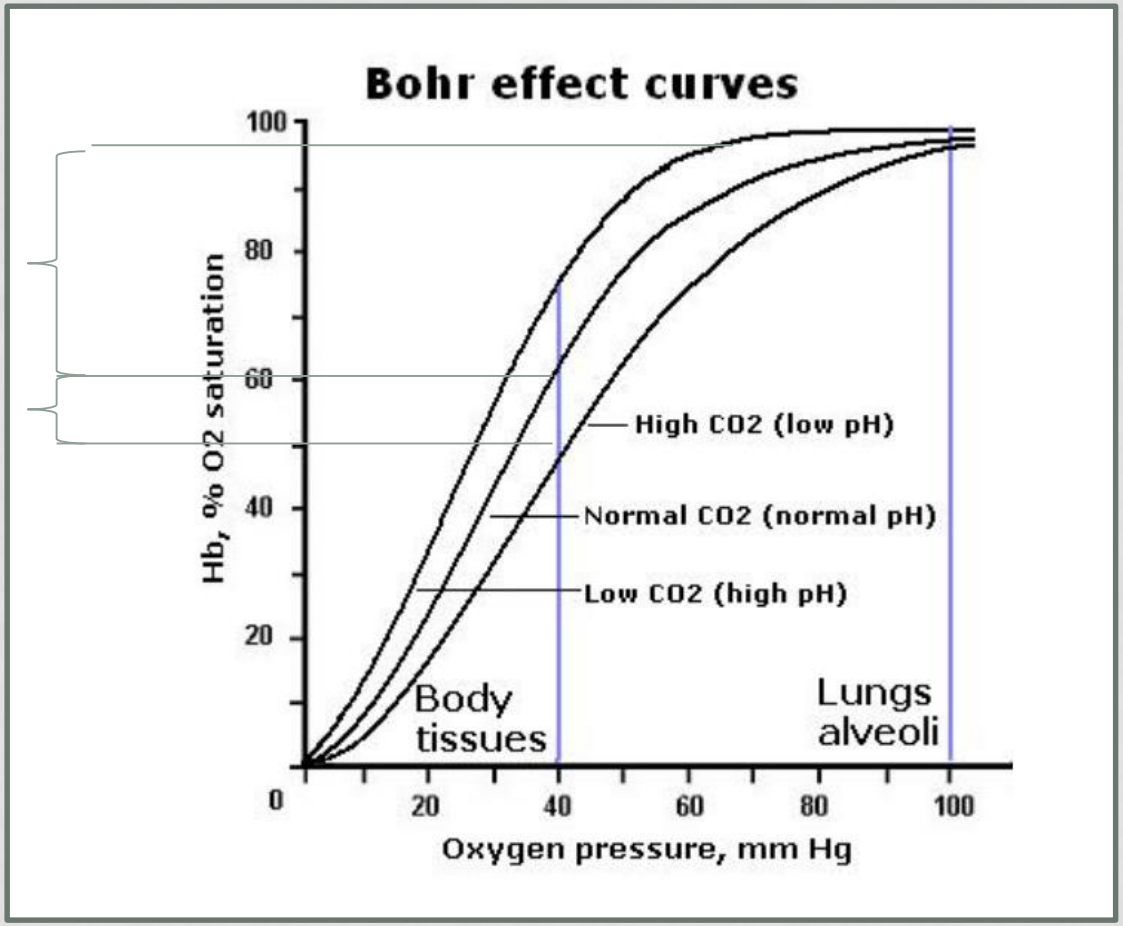
In lung, hemoglobin binds O_2 and liberates H^+ that reacts with HCO_3^- and forms CO_2 and H_2O . New HCO_3^- enter the red cells and Cl^- exit to keep neutral el. charge, arterial blood has more Cl^-)

- Oxyhemoglobin is a stronger acid (pK_A=6.2)

Hemoglobin: dissociation curve in acidosis

Normal pH: amount of released O_2

Acidosis: left shift: more O_2 is released

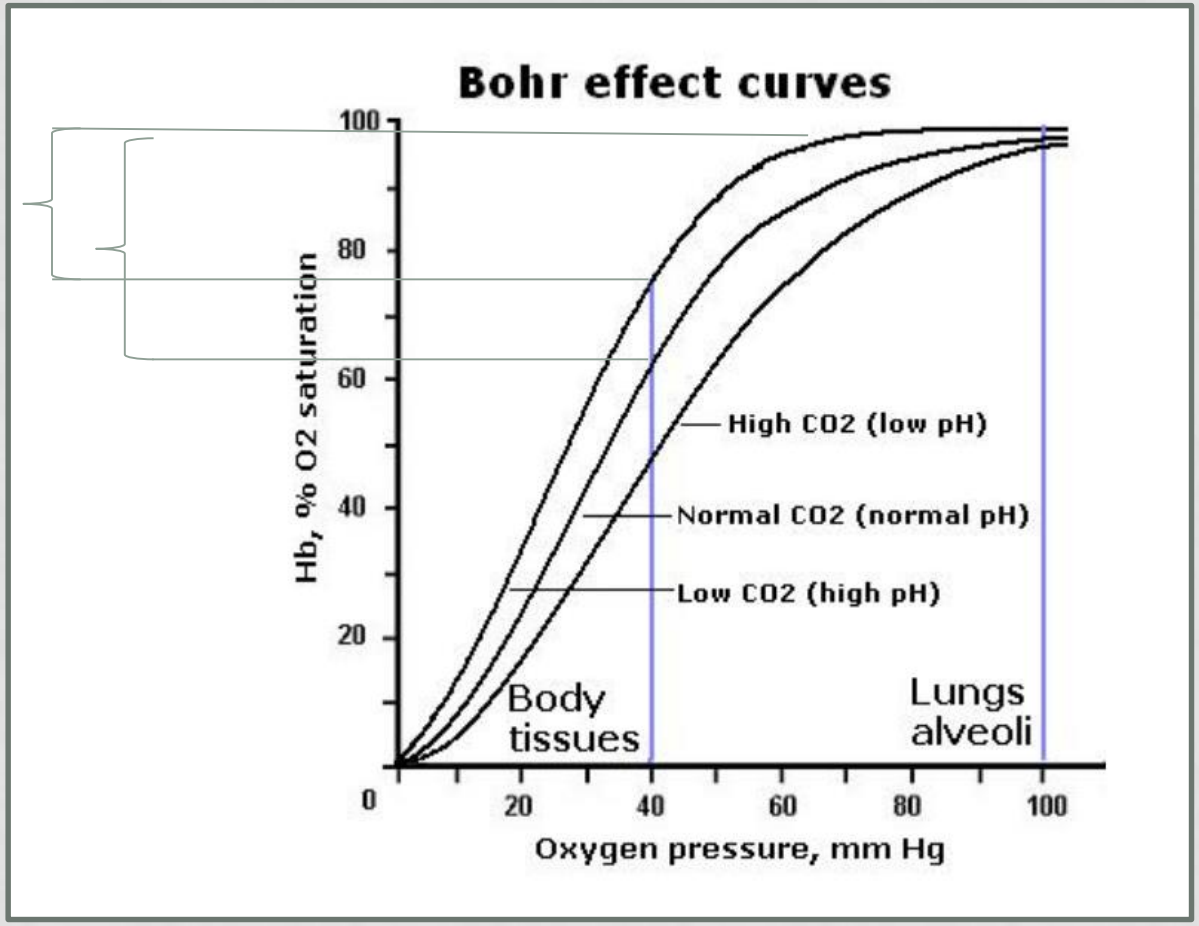


As acidosis progresses, 2,3-diphophoglycerate produced by glycolysis is going down and causes left shift

Hemoglobin: dissociation curve in alkalosis

Alkalosis: ↓ amount of released O_2 in tissue

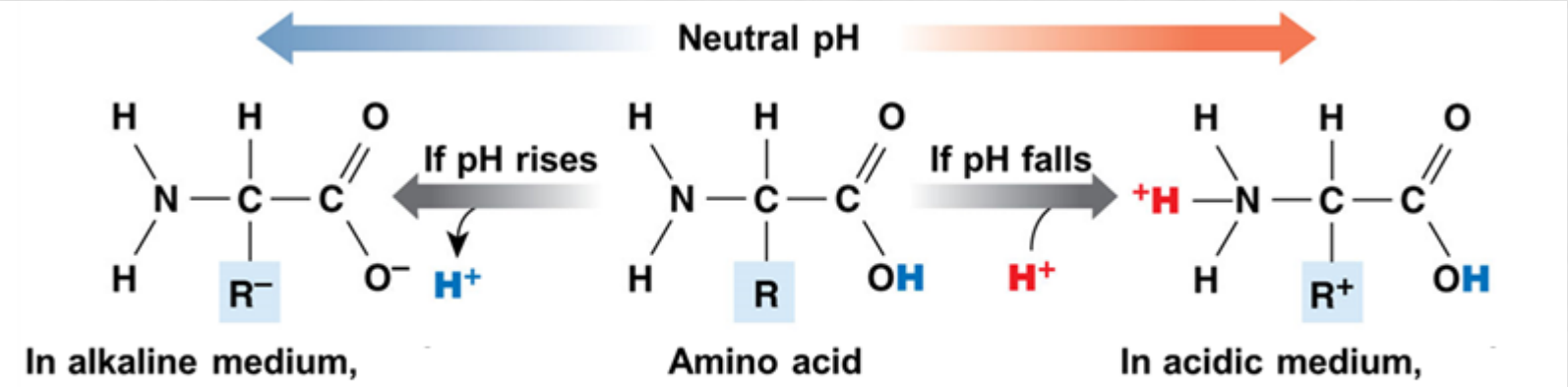
Normal pH: amount of released O_2



Alkalosis: ↑ affinity of Hb binds ↑ amount ofO_2 that is not released in tissue and the cells produce lactate that will compensate respiratory alkalosis

The proteins buffer

Hypoproteinemia (hypoalbuminemia) by itself causes a nonrespiratory ("metabolic") alkalosis.
Buffering by proteins accounts for two-thirds of the buffering power of the blood and most of the buffering within cells.



positively charged amino groups and negatively charged carboxyl groups.

The plasma proteins buffer accounts for 7% of all buffers

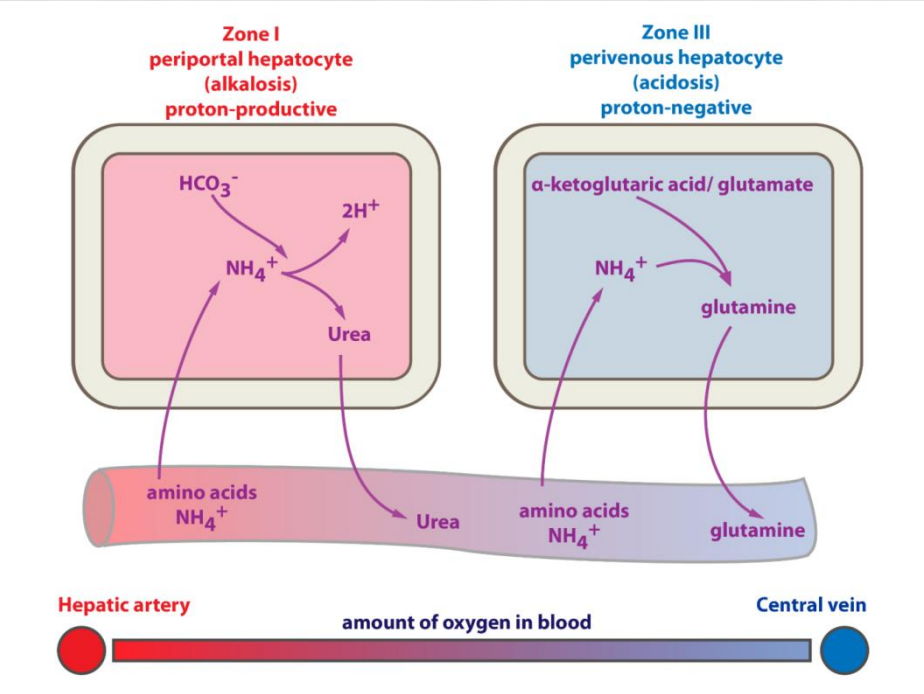
The liver: a normal source of urea

- Amino acids are deaminated (removal of an amino group) so that they can be processed into the Krebs/Citric Acid/TCA cycle.

- The amino group becomes ammonium and later undergoes the urea cycle to become urea in the liver.

    - $CO_2 + NH_4^+ \rightarrow$ urea + $2H^+$ (e.g. Sulphur containing amino acids and cationic amino acids $\rightarrow$ acidification)
    - $CO_2 + NH_4^+ \rightarrow$ urea without $2H^+$ (neutral amino acid)

The role of hepatocytes in acid base disturbance: production of urea or glutamine



- Urea is produced in less extent in acidosis. Hepatocytes synthetize more glutamine, because its production does not generate H^+.
    - NH_4^+ → **glutamine synthesis** → glutamine
- Glutamine is taken up by the kidneys. In the kidney **H^+** is **excreted as NH_4^+**

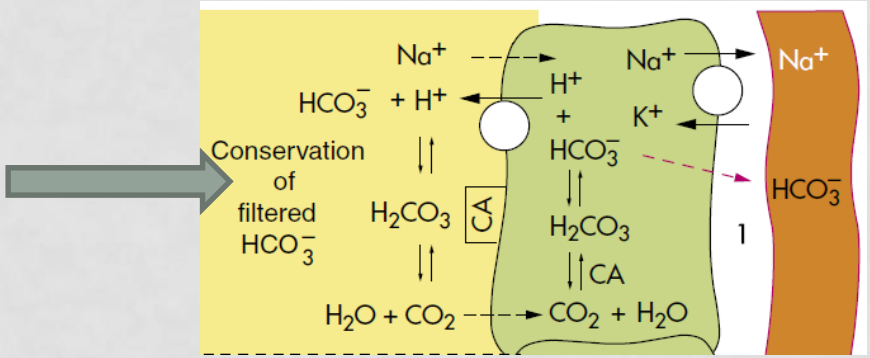
The role of hepatocytes in acid base disturbance: the consumption of excessive lactic acid

Lactic acid in the liver is firstly metabolized to pyruvic acid and then converted to glucose by gluconeogenesis. Both the release of lactic acid from the muscles and its metabolism into glucose is called the Cori cycle. This complete process results in the equivalent release of an [HCO3−] radical.

Kidney: normal function

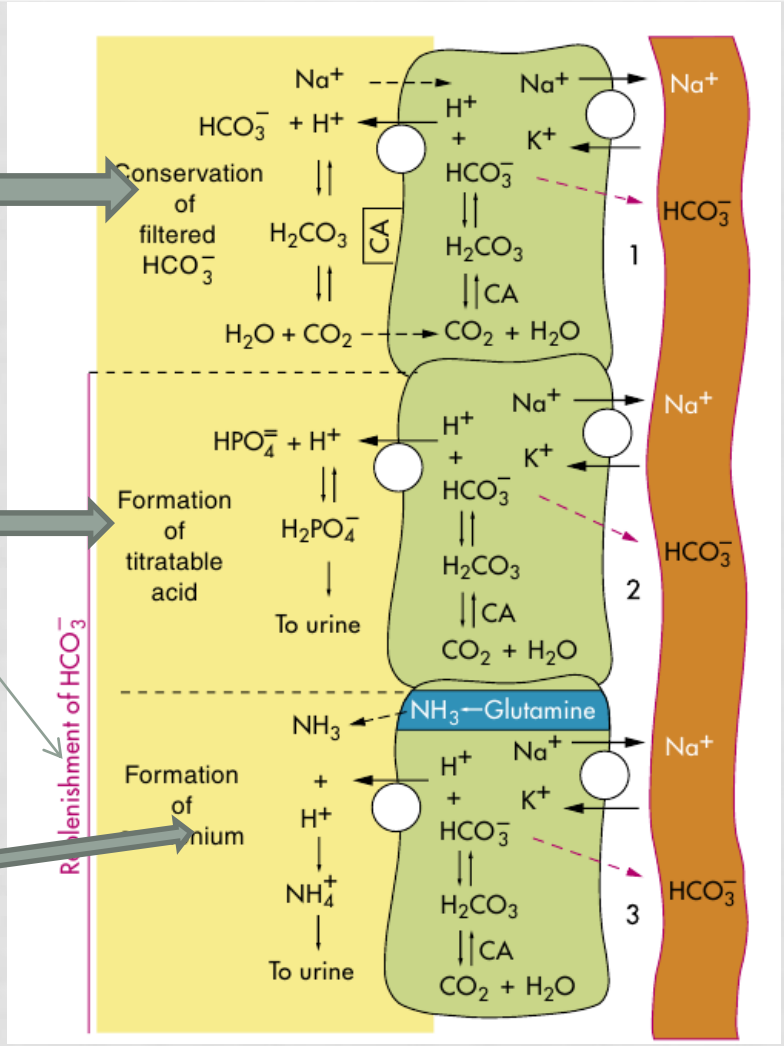
- The kidney excretes H^+: normal urine pH 4.5 – 8 and urine anion gap is negative.
- The kidney reabsorbed all filtered HCO_3^- .

In details: For every H^+ ion secreted into the tubular lumen (H^+ is actively exchanged for Na^+), HCO^-_3 enters the blood. 90% of filtered HCO^-_3 is reabsorbed in the form of CO_2 and HCO^-_3. H^+ normally do not contribute to the urinary excretion of acid or the addition of acid to the blood.



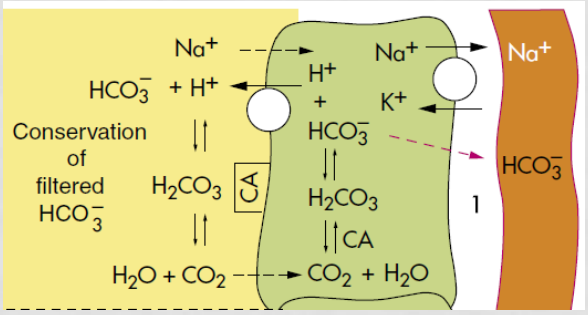
The role of kidney in acid base disturbance

1. Role of proximal tubule:
   1. HCO_3^- reabsorption
   2. Metabolize glutamine
2. Role of distal tubule:
   1. de novo HCO_3^- formation
   2. renal buffers:
   - The phosphate buffer accounts for 5% of all buffers
     - Important in urine due to acidic pH and higher concentration
   - NH_3/NH_4^+ Cl buffer

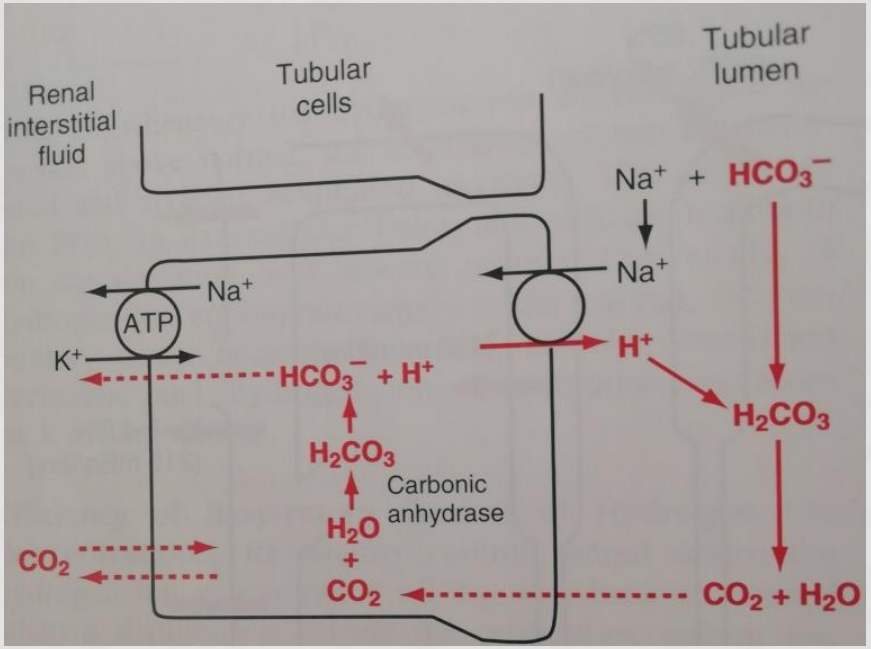


Proximal convoluted tubule: the effect of diuretics, hormones and genetic defect

- Acetazolamide: Inhibits carbonic anhydrase and thus blocks hydrogen ion secretion and reabsorption of sodium and bicarbonate“ leading to increased diuresis and side effects: hypokalemia, hyperchloremic, systemic acidosis, alkaline urine
- Na^+/H^+ pump: depend on available HCO^{-3}
  - arterial↑pCO_2stumulate Na^+/H^+ pump
  - ↑ angiotenzin II stumulate Na^+/H^+ pump
  - ↑ Parathormone decrease Na^+/H^+ pump (hyperparathyroidism leads to m. acidosis and alkaline urine)
- Proximal tubular acidosis: genetic change in Na+/H+ pump: not absorbed HCO^{-3}
  - IV with HCO^{-3} will not help because HCO^{-3} is no reabsorbed

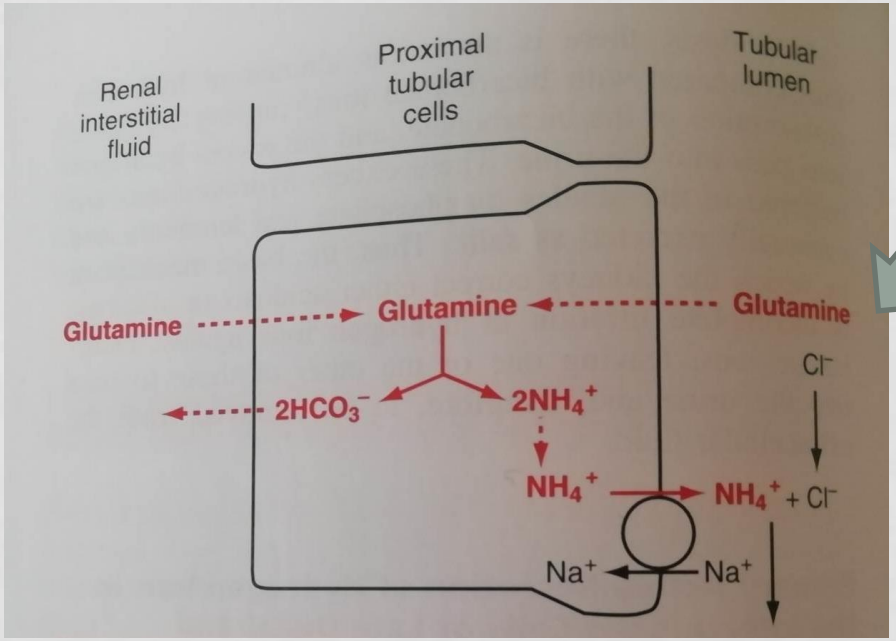


Proximal convoluted tubule: HCO_3^- reabsorption



Cellular mechanisms for (1) active secretion of hydrogen ions into the renal tubule; (2) tubular reabsorption of bicarbonate ions by combination with hydrogen ions to form carbonic acid, which dissociates to form carbon dioxide and water; and (3) sodium ion reabsorption in exchange for hydrogen ions secreted. This pattern of hydrogen ion secretion occurs in the proximal tubule, the thick ascending segment of the loop of Henle, and the early distal tubule.

Proximal convoluted tubule in metabolic acidosis: glutamine and NH_4^+ buffer

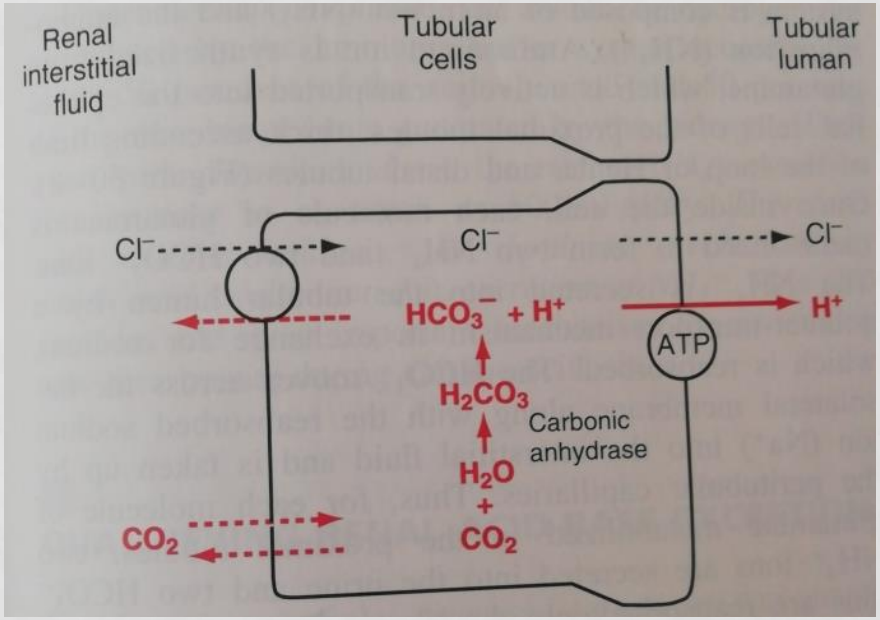


From the liver

Production and secretion of ammonium ion (NH_4^+) by proximal tubular cells. Glutamine is metabolized in the cell, yielding NH_4^+ and bicarbonate. The ammonium ion (NH_4^+) is actively secreted into the lumen by means of a sodium-NH_4^+ pump. For each glutamine molecule metabolized, two NH_4^+ are produced and secreted and two HCO_3^- are returned to the blood.

Intercalated cells of distal and collecting tubules:

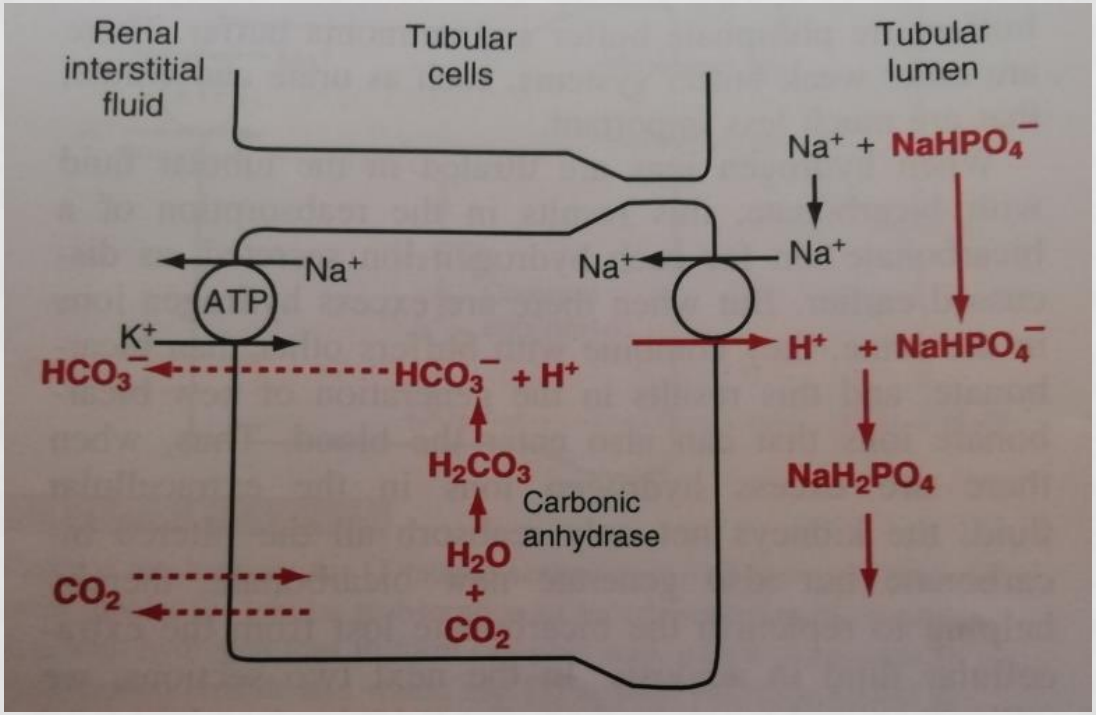
- De novo HCO_3^- production and secretion to blood
- Cl^- enter from blood to urine (antiporter with HCO_3^-)



- H^+ secretion to urine to be buffer with NH_3 or $H_2PO_4^-$ (the next 2 slides)

Primary active secretion of hydrogen ions through the luminal membrane of the epithelial cells of the distal and collecting tubules. Note that one bicarbonate ion is absorbed for each hydrogen ion secreted, and a chloride ion is passively secreted along with the hydrogen ion. This pattern of hydrogen ion secretion occurs in the intercalated cells of the late distal and collecting tubules.

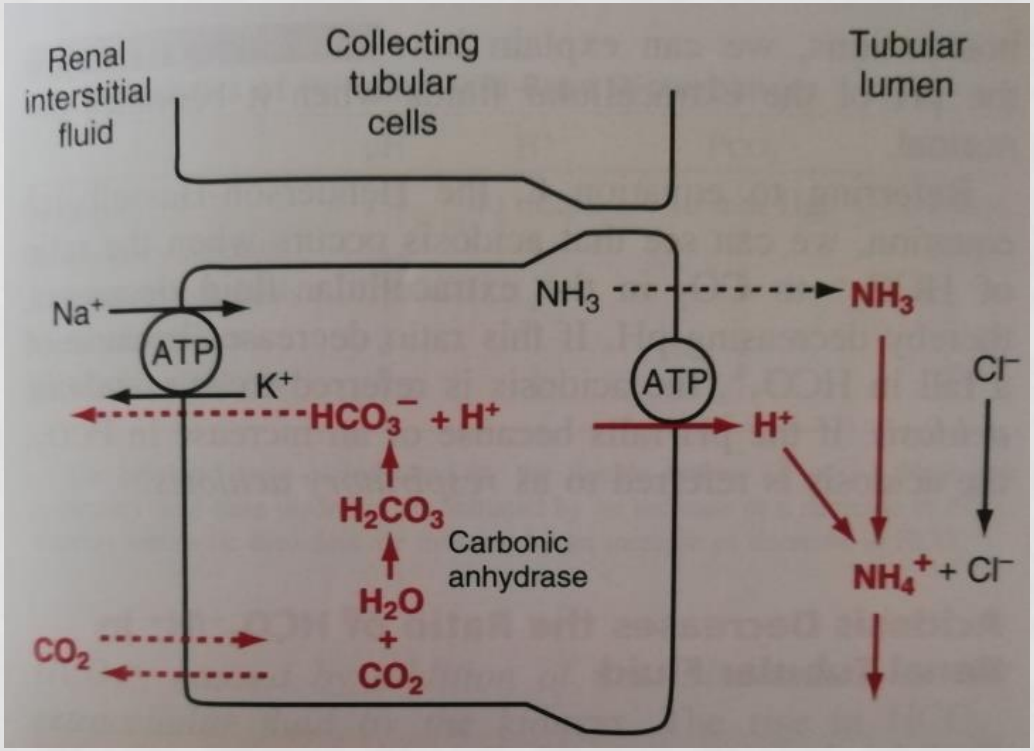
Distal tubule in metabolic acidosis: NaH_2PO_4 buffer



- H^+ secretion to urine to be buffered with $H_2PO_4^-$

Buffering of secreted hydrogen ions by filtered phosphate ($NaHPO_4^-$). Note that a new bicarbonate ion is returned to the blood for each $NaHPO_4^-$ that reacts with a secreted hydrogen ion.

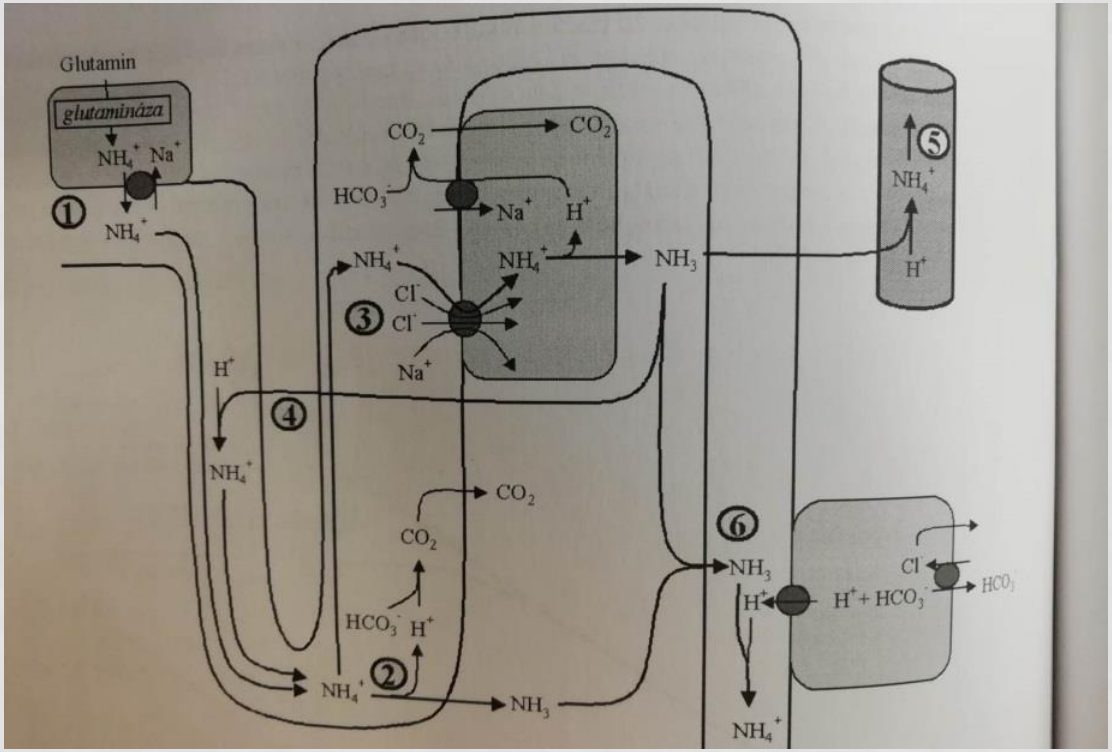
Collecting tubule in metabolic acidosis: NH_4^+ Cl buffer



Buffering of hydrogen ion secretion by ammonia (NH_3) in the collecting tubules. Ammonia diffuses into the tubular lumen, where it reacts with secreted hydrogen ions to form NH_4^+, which is then excreted. For each NH_4^+ excreted, a new HCO_3^- is formed in the tubular cells and returned to the blood.

- H^+ secretion to urine to be buffered with NH_3

The recycling of NH_3



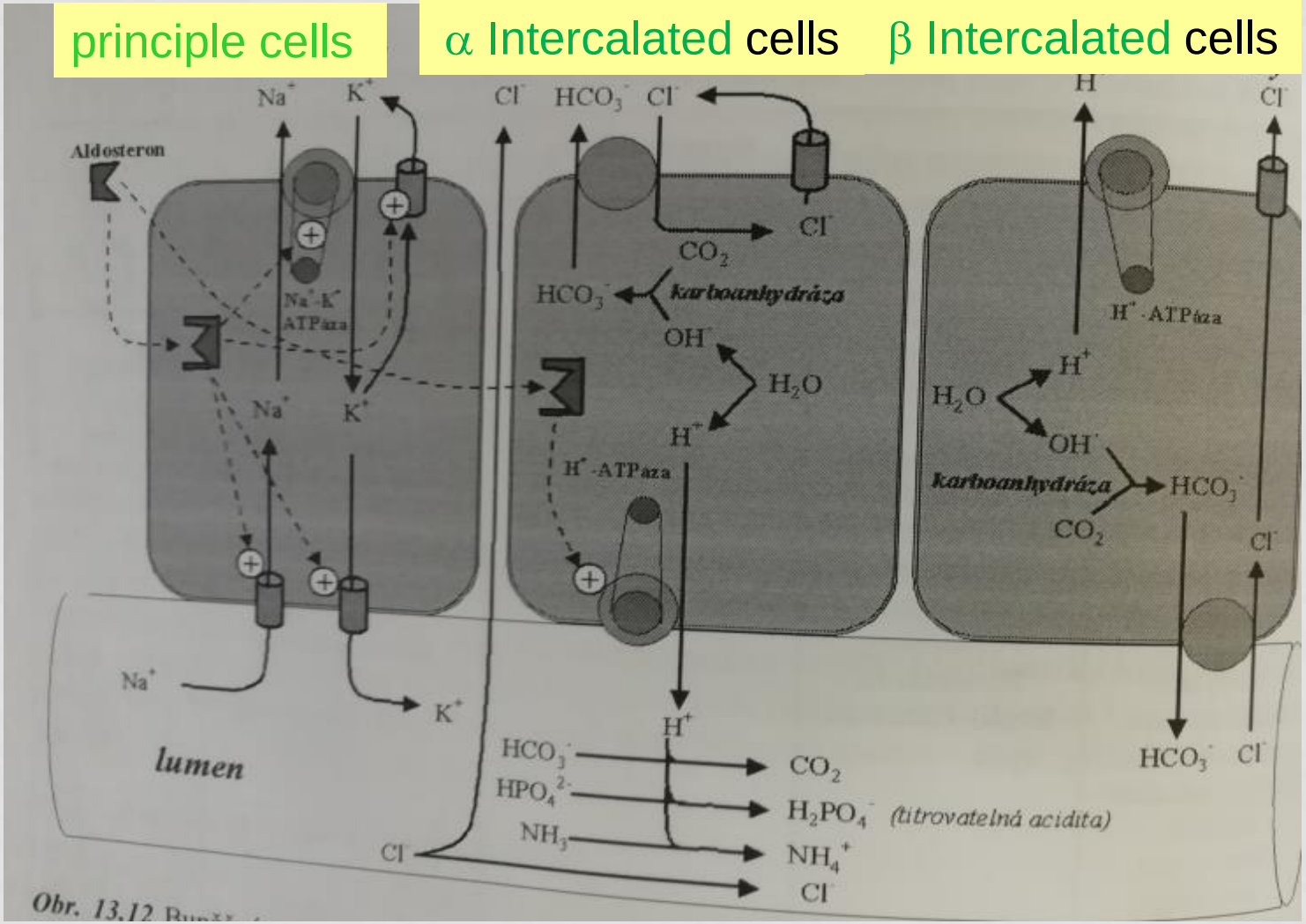
NH_4^+ from glutamine:

- Reabsorbed via $Na^+2Cl^-NH_4^+$ in ascending loop
- Release NH_3 that diffuse to interstitium and partially diffuse to descending loop where NH_3 binds H^+.

The interstitium releases as much NH_3 as H^+ is in collecting tubules

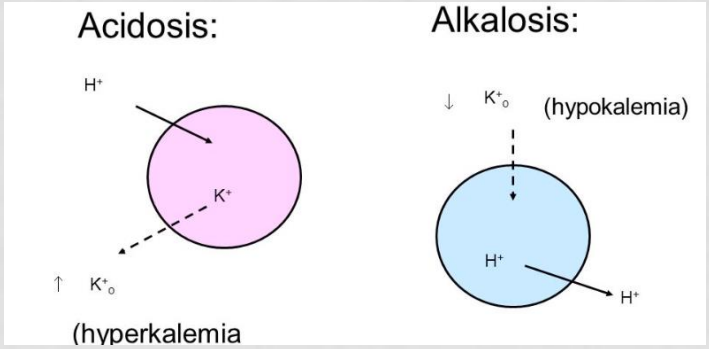
The distal tubule: a role of aldosterone

A intercalated cells secrete H^+ into the tubular lumen and HCO_3^- into blood in exchange for Cl^-



Interchangeable anions and cations

- Interchangeable K^+ and H^+ (two—way mechanism)
  - Acidosis can be associated with hyperkalemia but may lead to hypokalemia due to the increased excretion.
  - Alkalosis can be associated with hypokalemia
- Interchangeable Cl^- and HCO_3^-
  - Acidosis can be associated with increased Cl^- (hyperchloremic / normal anion gap) or normal Cl^- (normochloremic / high anion gap)
  - Alkalosis can be hypochloremic



Interchangeable ions in acidosis: K^+ and H^+

- In high anion gap acidosis (e.g. excess of organic acid in lactic acidosis or diabetic acidosis), a proton enters cell with anions and both pumps H^+/Na^+ and Na^+/K^+ work, leading to hypokalemia that can be supported by osmotic diuresis or the effect of insulin in diabetic acidosis.
- In normal anion gap acidosis, both intra and extracellular concentrations of H^+ are elevated and inhibit both pumps: hyperkalemia

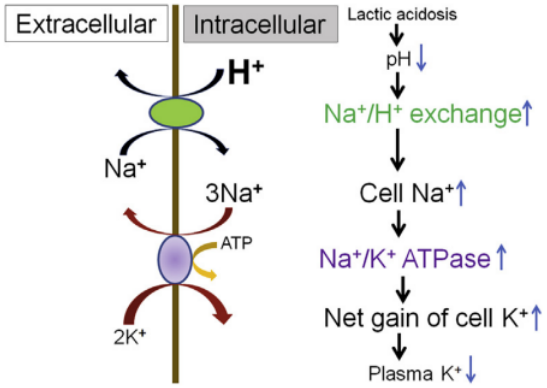
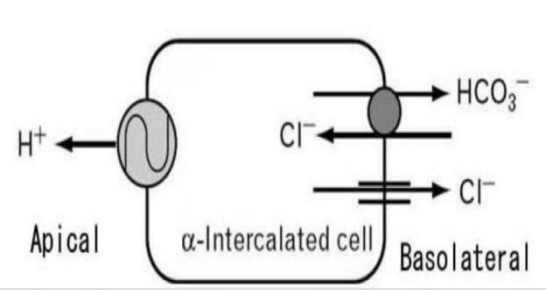


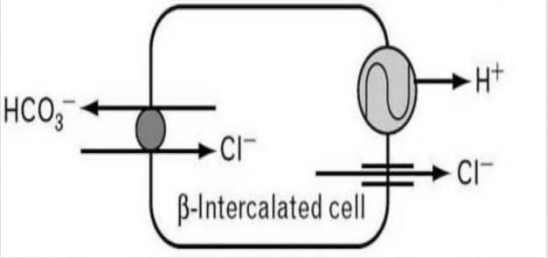
Figure 1. Schematic of effects of lactic acidosis on plasma K^+. Elevation of lactate, an organic acid, causes lactic acidosis, which results in reduction of intracellular pH. The low pH in turn activates the Na^+/H^+ exchange in the cell, which pushes H^+ out in exchange for Na^+. The influx of Na^+ intracellularly then activates the Na^+/K^+-ATPase, which results in intracellular shift of K^+, leading to hypokalemia. Adapted from Figure 4 in Aronson PS, Giebisch G. Effects of pH on potassium: new explanations for old observations. *J Am Soc Nephrol.* 2011;22:1981–1989.[2] Copyright © 2011 by the American Society of Nephrology.

- Hypoaldosteronism: ↑ plasma K^+ is not excreted: hyperkalemia (↓aldosterone does not stimulate its excretion)
- RTA: ↑ plasma K^+ is excreted as negative luminal charge (↓urinary acidification) support K^+ excretion: hypokalemia

Interchangeable ions: HCO_3^- and Cl^-

- Interchangeable Cl- and HCO3-
  - Acidosis can be associated with increased Cl^- (hyperchloremic= normal anion gap) or normal Cl^- (normochloremic= high anion gap)
  - Alkalosis can be hypochloremic
- Red cells:
- Nephron:
  - Intercalated cells A: basolateral side
  - Intercalated cells B: apical side





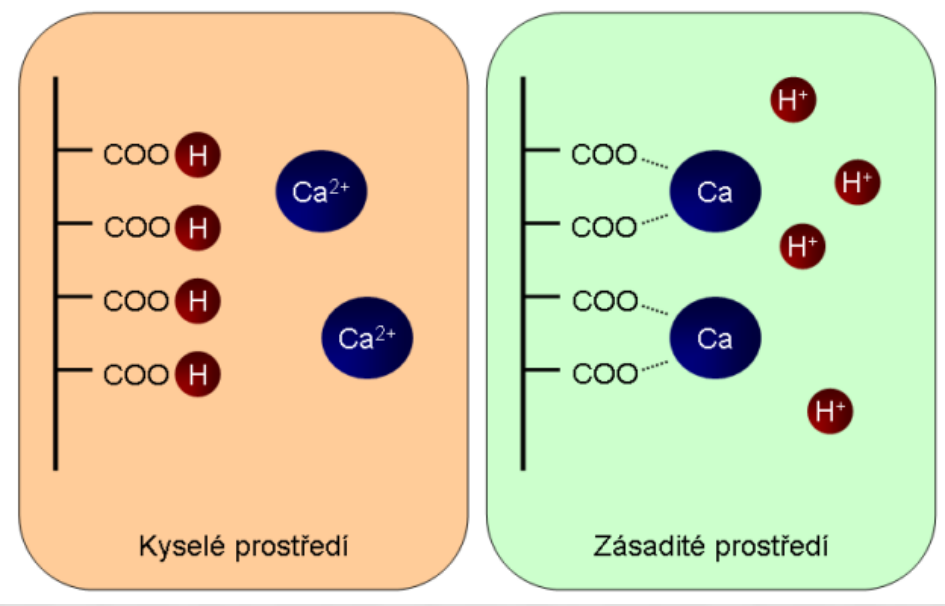
Ca^{+} Bone buffer: Calcium

- Most Ca^{2+} (99%): in bone as hydroxyapatite ($Ca_5(PO_4)_3(OH)$)
- Small fraction of Ca^{2+} circulating within the intracellular (primarily in the mitochondria) and extracellular (in the blood):
    - 50% is bound to plasma proteins, (primarily albumin) and
    - 40% is in the free or ionized form (5.5 to 5.6 mg/dl).

Ca^{+}

The ionized calcium

In states of acidosis, levels of ionized Ca^{2+} increase.
When alkalosis develops, with an increase in pH, the amount of protein-bound calcium increases and the physiologically active, ionized Ca^{2+} level decreases. The decreased concentration of ionized Ca^{2+} may be great enough to cause symptoms of hypocalcemia, such as tetany.



3. Clinical Approach to Acid-Base Disturbances

Astrup

It refers to 4 parameters that assess acid-base disturbance:

pH

pO_2

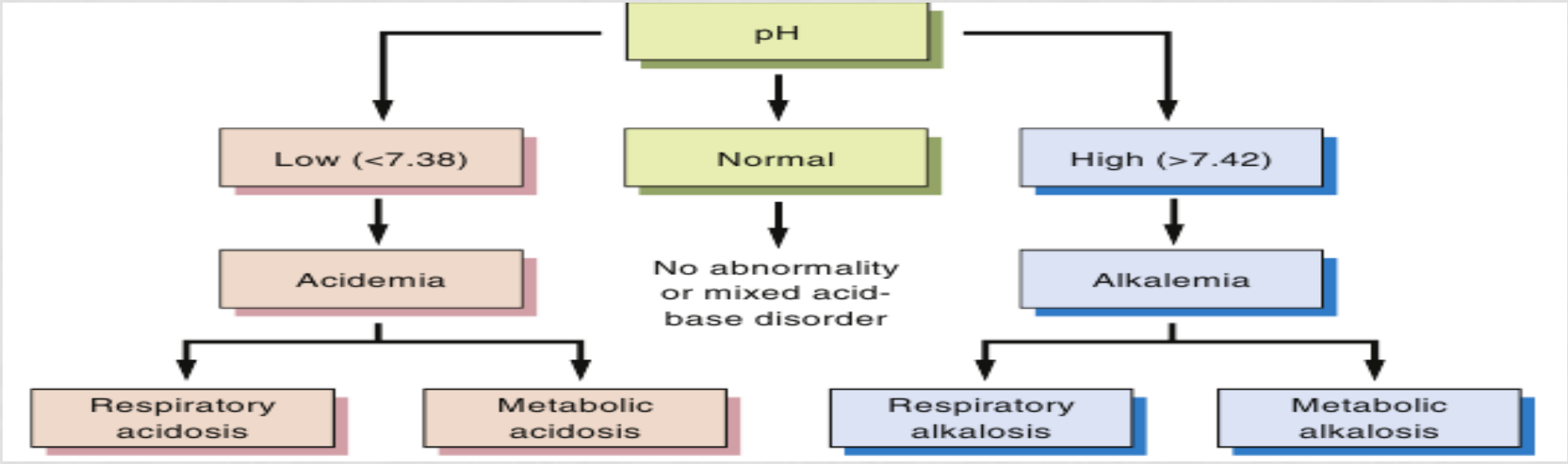
pCO_2

HCO_3^-



Professor Poul Astrup

Astrup



↑pCO_2 (normal = 40 mmHg (33-44))

Cause: lung diseases

↓HCO_3^- (normal = 24 mmol/L (22-28)

Cause: renal, liver or other diseases

↓ pCO_2 (normal = 40 mmol/L (33-44))

Cause: lung diseases

↑HCO_3^- (normal = 24 mmol/L (22-28))

Cause: renal, liver or other diseases

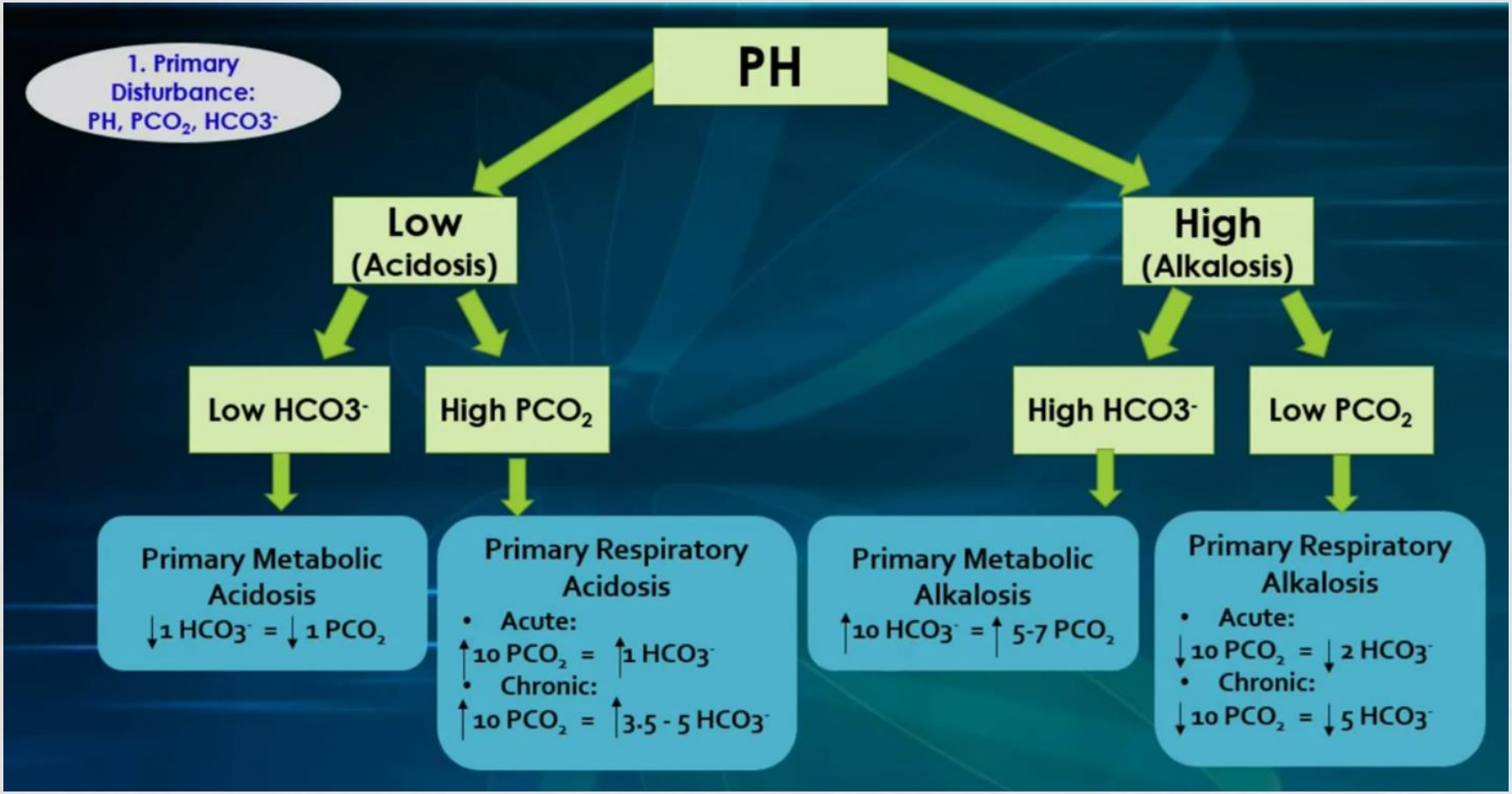
Henderson-hasselbalch equation: compensation

$$pH = 6.1 + \log\left[\frac{HCO_3^-}{(0.03)(PCO_2)}\right]$$

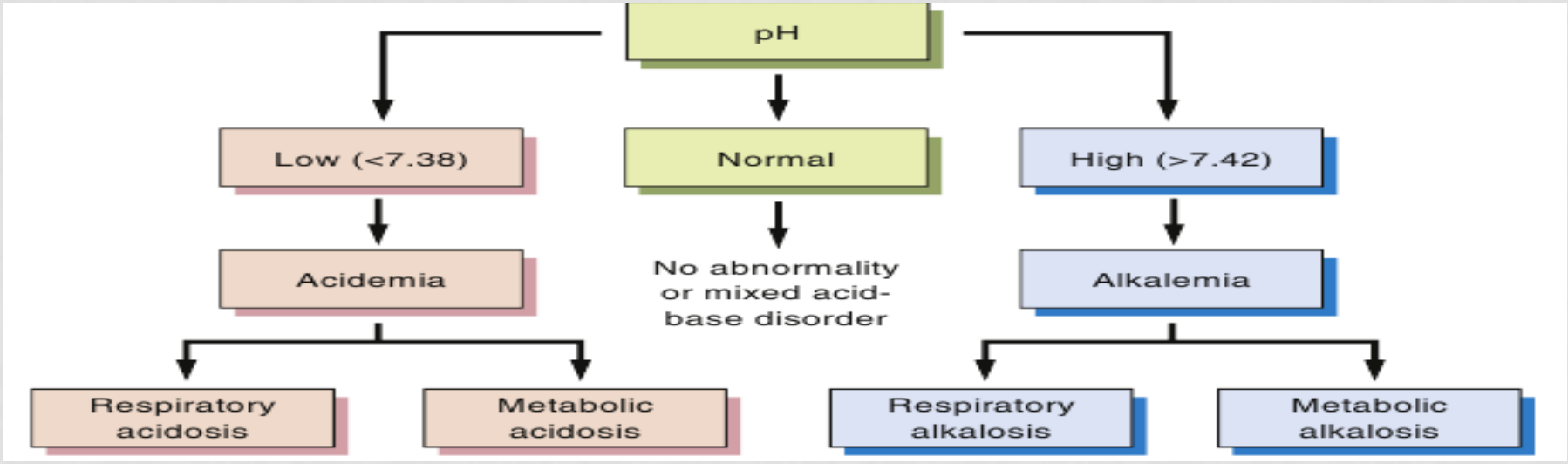
- 6.1: Dissociation constant
- 0.03: Solubility coefficient for CO_2

- There is a simple fraction in the equation.
- It means that if a numerator (HCO_3^-) is increased, a denominator (CO_2) also must increase to keep normal (physiological) pH.
- It means that if a numerator (HCO_3^-) is decreased, a denominator (CO_2) also must decrease to keep normal (physiological) pH.

All simple AB disturbances with their compensations together



Respiratory component with/without compensation: acidosis



$\uparrow pCO_2$

Cause: lung diseases

$\downarrow HCO_3^-$

Cause: renal, liver or other diseases

$\downarrow pCO_2$

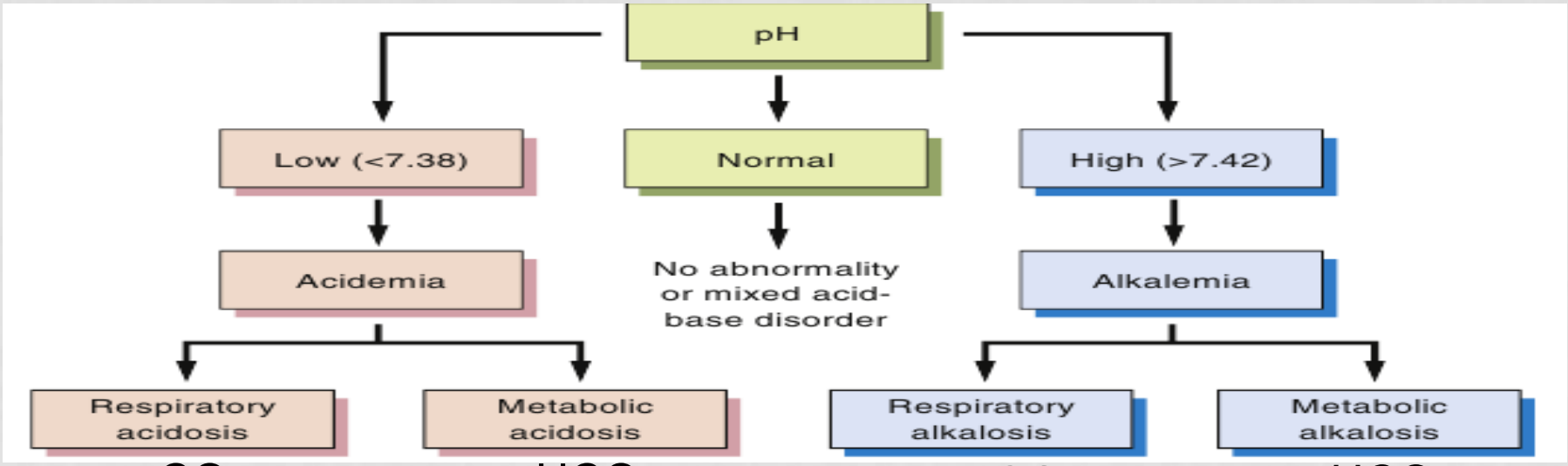
Cause: lung diseases

$\uparrow HCO_3^-$

Cause: renal, liver or other diseases

- Without compensation:
  - Hypoventilation
  - Pickwickian syndrome
- With compensation after (3-5 days for kidney) : COPD

Respiratory component with/without compensation: alkalosis



$\uparrow pCO_2$

Cause: lung diseases

$\downarrow HCO_3^-$

Cause: renal, liver or other diseases

$\downarrow pCO_2$

Cause: lung diseases

$\uparrow HCO_3^-$

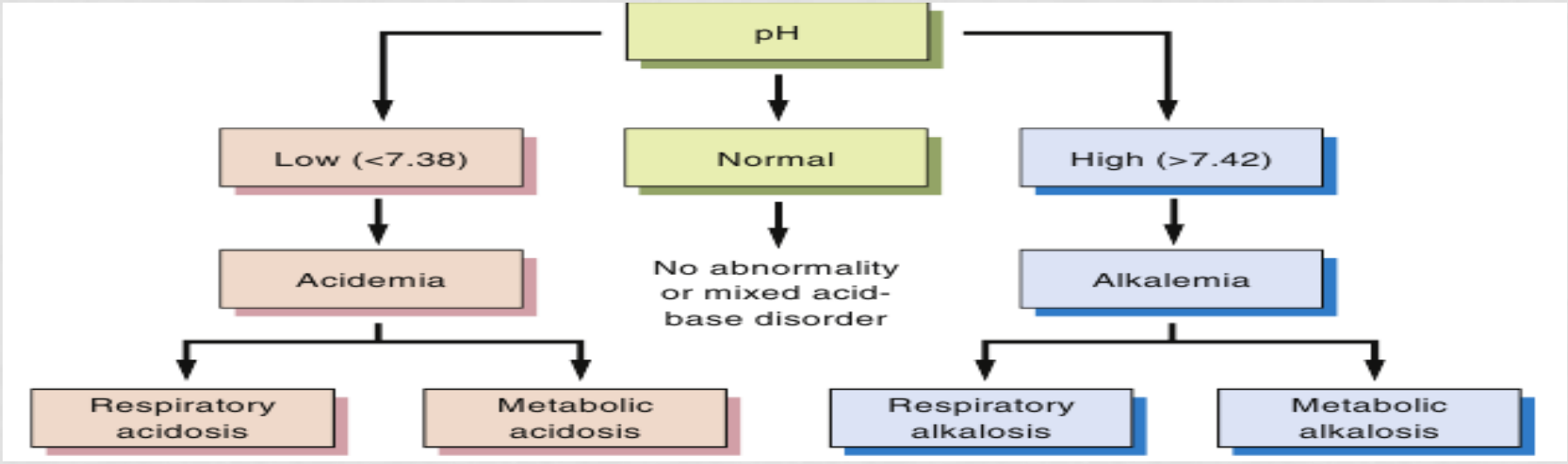
Cause: renal, liver or other diseases

Without compensation:

- Hyperventilation
- staying in high altitude
- anxiety attack

With compensation after (3-5 days for kidney)

Metabolic component with/without compensation: acidosis



$\uparrow pCO_2$

Cause: lung diseases

$\downarrow HCO_3^-$

Cause: renal, liver or other diseases

$\downarrow pCO_2$

Cause: lung diseases

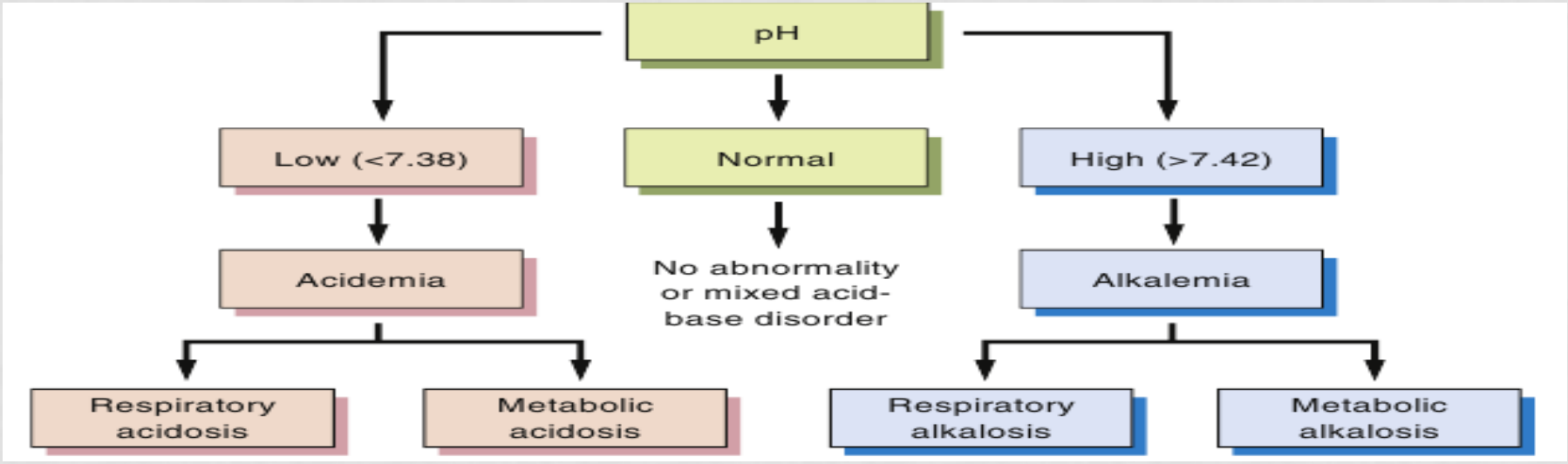
$\uparrow HCO_3^-$

Cause: renal, liver or other diseases

With compensation as the healthy lung can compensate rapidly (6-12 hours)

- high anion gap = normochloremia
- normal anion gap= hyperchloremia

Metabolic component with/without compensation: alkalosis



$\uparrow pCO_2$

Cause: lung diseases

$\downarrow HCO_3^-$

Cause: renal, liver or other diseases

$\downarrow pCO_2$

Cause: lung diseases

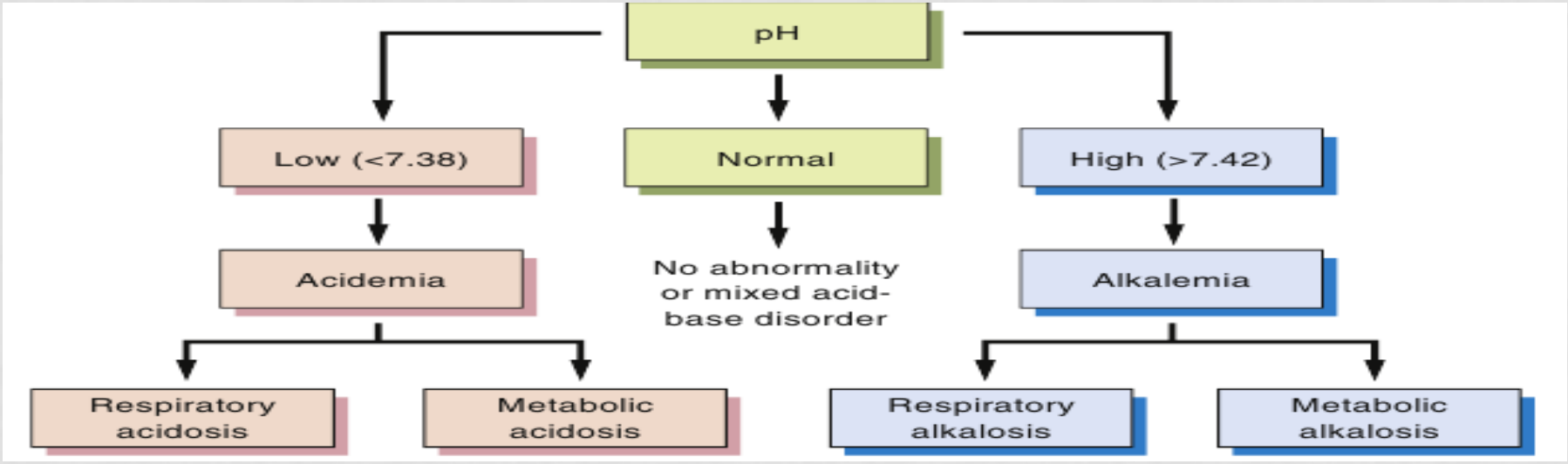
$\uparrow HCO_3^-$

Cause: renal, liver or other diseases

With compensation after 6-12 hours for lung

- Alkalosis with high Cl^- loss in urine
- Alkalosis with lower Cl^- loss in urine

Metabolic and respiratory component with/without compensations: summary



$\uparrow pCO_2$

Cause:
lung diseases

- Without compensation:
  - Hypoventilation
  - Pickwickian syndrome
- With compensation after (3-5 days for kidney) : COPD

$\downarrow HCO_3^-$

Cause: renal, liver or other diseases

With compensation as the healthy lung can compensate rapidly (6-12 hours)

- high anion gap = normochloremia
- normal anion gap= hyperchloremia

$\downarrow pCO_2$

Cause:
lung diseases

Without compensation:

- Hyperventilation
- staying in high altitude
- anxiety attack

With compensation after (3-5 days for kidney)

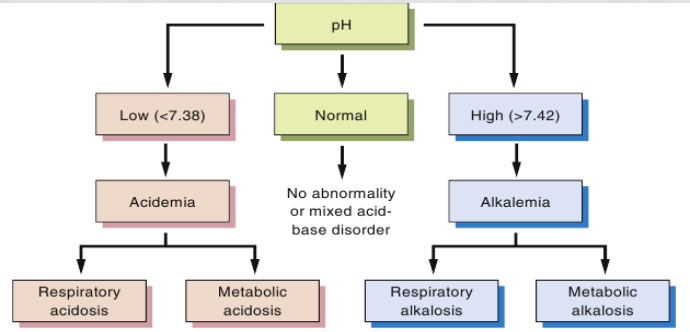
$\uparrow HCO_3^-$

Cause: renal, liver or other diseases

With compensation after 6-12 hours for lung

- Alkalosis with high Cl^- loss in urine
- Alkalosis with lower Cl^- loss in urine

Compensation and correction



Powerful compensation of respiratory acidosis takes time:

- Kidney: ↑ pCO_2 leads to excess of H^+ over HCO_3^- in the tubular fluid and the ↑ H^+ induce Na^+/H^+ pump leading to intensive renal reabsorption of HCO_3 and mainly renal formation of new HCO_3^-
- Liver: glutamine over urea is synthetized

Compensation of metabolic acidosis:

- Fast and less powerful lung: decreases pCO_2 (acids) by hyperventilation

Correction of metabolic acidosis:

- ↓ filtration of HCO_3^- leads to an excess of H^+ over HCO_3^- in the renal tubular fluid and the ↑ H^+ induce Na^+/H^+ pump leading to renal formation of new HCO_3^- if met. acid is not due to renal diseases
- ↑ **NH_4^+Cl and NaH_2PO_4 in urine**

Maintenance of the acid-base balance: timetable

1) Chemical buffering systems
   - **“Quick” regulation**. Buffers react immediately (ms). Capacity of buffers is not indefinite that is why chemical buffers act only in the short-term. Chemical buffering systems deal with pH deviations in common metabolism.
2) Respiratory system “**Intermediate**”
   - Respiration starts to react in 1-3 minutes. Respiratory system regulates PaCO2, max activation after 12 hours
3) Kidneys (**“Slow”**)

   React in hours-days, max activation after 3-5 days.
4) Myocardium
   - Influences acid-base balance through lactate and ketone bodies oxidation.

Metabolic acidosis: „ANION- GAP“

$AG = (Na^+ + K^+) - [(HCO_3^- + Cl^-)]$

AG = 12 +/- 2 mmol/L

The AG value represents the difference between measured cations and anions, ie, the presence of anions in the plasma that are not routinely measured including:

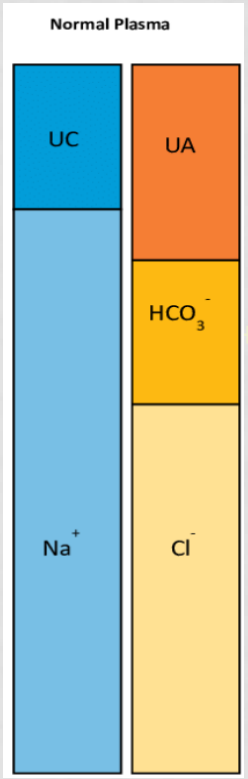
- Lactate (anion of lactic acid) in lactic acidosis
- Phosphates and sulfates in last stage of renal failure
- Keto bodies (acetoacetic acid, hydroxybutyric acid and acetone) in ketoacidosis
- Other anions in ingestion of certain toxins.
- Albumins

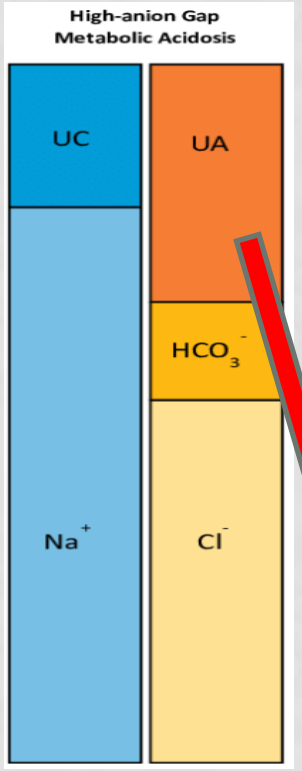
Metabolic acidosis: with ↑ AG or normal AG

Normal plasma

High Anion Gap without hyperchloremia

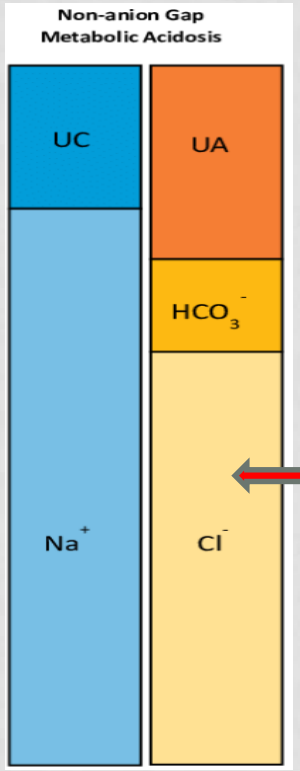
Normal Anion Gap with Hyperchloremia





High AG acidosis is characterized by

- a lowered HCO_3^- concentration, which is not counterbalanced by an equivalent increase in plasma Cl^- concentration.
- Accumulation of strong acid anions: lactate anion or keto bodies



A normal AG acidosis is characterized by a lowered HCO_3^- concentration, which is counterbalanced by an equivalent increase in plasma Cl^- concentration due to ↑ renal reabsorption

UA: Unmeasured cations / anions

1 Buffer base and Buffer excess

“Buffer base” (BB) is the sum of bicarbonate and non-bicarbonate buffer anions (ie, HCO_3^- + K*albumins +K2*hemoglobin).

Base excess is the quantity of base (HCO_3^-, in mEq/L) that is above or below the normal range of buffer base in the body (22 -28 mEq/L).

2 Buffer base and Buffer excess

"BE" is defined as the amount of acid or base that must be added in vitro to a sample of whole blood to restore the pH to 7.40 while $PaCO_2$ is kept constant at 40 mmHg (5.33 kPa). Oxygenated blood at $PaCO_2$ 40 mmHg (5.33 kPa) and Hb 150 g/L has a BE of zero

Normal Base excess is between -3 and +3mEq/L.
Negative base excess (e.g. BE –8 mmol/L) indicates metabolic acidosis
Abnormally positive base excess (e.g. BE +8 mmol/L) indicates metabolic alkalosis.

High Anion Gap without hyperchloremia: causes

Causes

- Lactic acidosis (buildup of lactate)
- Ketoacidosis (DM I type, starvation, alcoholic ketoacidosis)
- Last stage of chronic renal failure (accumulation of sulfates, phosphates, uric acid)
- Intoxication:
  - Organic acids (salicylates, ethanol, methanol, formaldehyde, ethylene glycol, paraldehyde, INH, toluene)
  - Sulfates, metformin (Glucophage)
- Massive rhabdomyolysis

Delta Ratio

- Delta Ratio is AG scale
- Calculation: (AG – 12)/(24 - [HCO_3^-])
  - < 0.4 due to NAGMA (normal anion gap metabolic acidosis)
  - 0.4 - 0.8 due to a mixed NAGMA + HAGMA
  - 0.8 - 2.0 due to a pure HAGMA (high anion gap metabolic acidosis e.g. Lactic acidosis)
  - >2.0 due to a mixed HAGMA + metabolic alkalosis (or pre-existing compensated respiratory acidosis) (e.g. COPD)

High Anion Gap without hyperchloremia: mnemonics

- **KULT** is probably the easiest of the mnemonics to use (Ketones, Ureamia, Lactate, Toxins).

- **MUDPILES**
  - M-Methanol poisoning
  - U-Uremia
  - D-Diabetic Ketoacidosis
  - P-Paraldehyde
  - I-Infection, Iron, Isoniazid
  - L-Lactic acidosis
  - E-Ethylene Glycol
  - S-Salicylates

- **GOLDMARK**
  - G — glycols (ethylene glycol & propylene glycol)
  - O — oxoproline, a metabolite of paracetamol
  - L — L-lactate, the chemical responsible for lactic acidosis
  - D — D-lactate causes of anion gap metabolic acidosis include
  - M — methanol
  - A — aspirin
  - R — renal failure
  - K — ketoacidosis, ketones generated from starvation, alcohol, and diabetic ketoacidosis

Normal Anion Gap with hyperchloremia: causes

- GIT
- kidney

Normal Anion Gap with hyperchloremia: 2 causes

urine anion gap AG = ([Na^+] + [K^+]) - [Cl^-]

- If kidney function is normal, urine anion gap negative.
- Bicarbonate loss from body fluids through the GI tract, with subsequent relative chloride retention: hyperchloremic metabolic acidosis (e.g. Longstanding diarrhea, vomiting the content from deeper in the gastrointestinal tract). Negative urine gap as the kidney compensate by ↑NH_4^+ excretion that are accompanied by ↑Cl^- excretion

- If kidney function is not normal, urine anion gap is 0 or positive
- ↓ HCO_3^- reabsorption or ↓H^+ excretion in the kidney with subsequent Cl^- retention: hyperchloremic metabolic acidosis (e.g. renal tubular acidosis (RTA) .Positive or 0 urine gap as the kidney reabsorb Cl^-: ↓ urinary acidification increases Cl^- reabsorption

Diarrhea: normal AG in blood and urine

Bicarbonate loss from body fluids through the GI tract , with subsequent relative chloride retention: hyperchloremic metabolic acidosis with normal AG in the blood: AG = ([Na^+] + [K^+]) - [Cl^-] + HCO_3^-]

Normal (negative) urine gap as the kidney compensate by ↑NH_4^+ excretion that are accompanied by ↑Cl^- excretion: urine anion gap AG = ([Na^+] + [K^+]) - [Cl^-]

Hypokalemia may be present: the stool has ↑K^+ and low Na^+ and low Cl^-

Renal tubular acidosis (RTA)

- Proximal renal tubular acidosis (type II): with hyperchloremia and hypokalemia. Rare
- Distal renal tubular acidosis (type I.): with hyperchloremia and hypokalemia. Rare
- Renal tubular acidosis (type III): with hyperchloremia and normokalemia but later normochloremia, and hyperkalemia. It is combination of type I and II. Rare
- Generalized renal tubular acidosis (type IV): with hypochloremia and hyperkalemia

↓ amount of blood HCO_3^- and↑ urinary HCO_3^- leads to accumulation of Cl^- and ↑ K^+ excretion (due to more negative charge in tubular lumen)

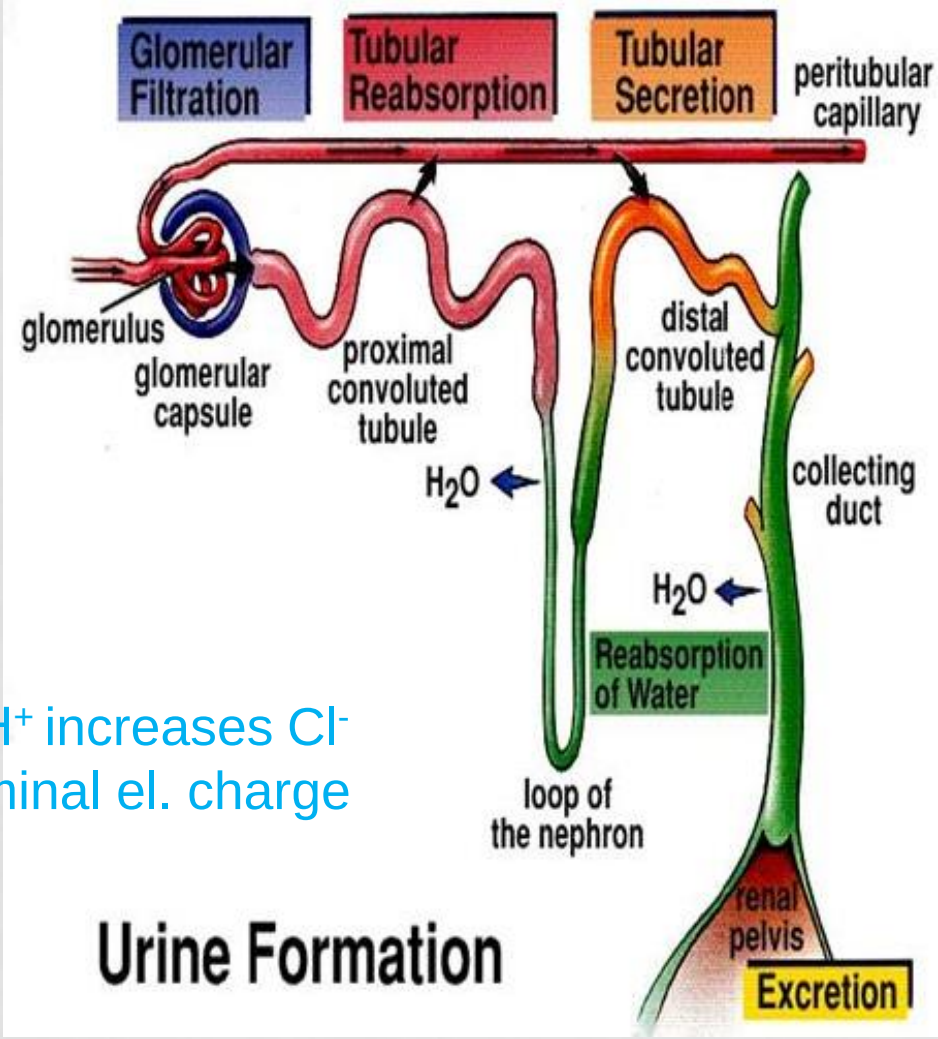
The renal interstices is damaged by the inflammation or decreased glomerular filtration: ↑ urinary NH_4^+ reabsorbed to blood

All RTA with normal AG

Renal tubular acidosis type III. the renal interstitium is damaged by the inflammation or decreased glomerular filtration

Proximal renal tubular acidosis (type II.) is caused by a failure of the proximal tubular cells to reabsorb bicarbonate from the urine.

All RTA: ↓ urinary H^+ increases Cl^- reabsorption as luminal el. charge is more negative



Distal renal tubular acidosis (type I.) is caused by a failure of the distal tubular cells to secrete H+ into lumen of nephron.

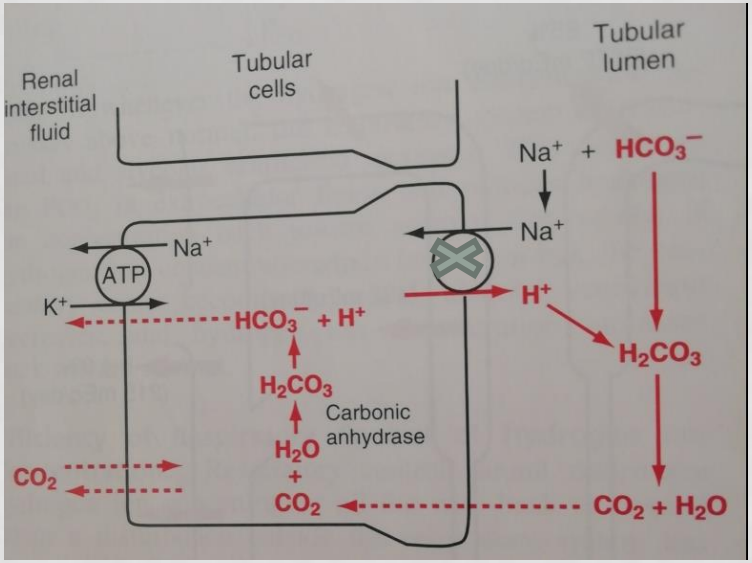
Renal tubular acidosis Type 4 absolute hypoaldosteronism or aldosterone insensitivity (principal cells of collecting ducts)

Type II: proximal renal tubular acidosis

is caused by a failure of the proximal tubular cells to reabsorb bicarbonate from the urine. There is defect in H^+/Na^+ antiporter that is associated with HCO_3^- reabsorption. Type II is associated with hyperchloremia and hypokalemia

- Hyperchloremia: To keep electric equilibrium, Cl^- are accumulated leading to hyperchloremia (*).
- Hypokalemia: Excess of urinal bicarbonate in distal tubule increases excretion of K^+ leading to hypokalemia: Excessive urinal bicarbonates bind Na^+ that is reabsorped by principal cells and bicarbonates negative charge stimulate K^+ excretion

Proximal convoluted tubule: Type II RTA



Type II is a defect in H+/Na+ antiporter that is associated with HCO_3^- reabsorption.

Cellular mechanisms for (1) active secretion of hydrogen ions into the renal tubule; (2) tubular reabsorption of bicarbonate ions by combination with hydrogen ions to form carbonic acid, which dissociates to form carbon dioxide and water; and (3) sodium ion reabsorption in exchange for hydrogen ions secreted. This pattern of hydrogen ion secretion occurs in the proximal tubule, the thick ascending segment of the loop of Henle, and the early distal tubule.

Type I: distal renal tubular acidosis

is caused by a failure of the distal tubular cells to secrete H+ into lumen of nephron. Low H+ in distal lumen decreases bicarbonate reabsorption. Type II is associated with hyperchloremia and hypokalemia

- Hyperchloremia: To keep electric equilibrium, Cl^- are accumulated leading to hyperchloremia (normal AG).
- Hypokalemia: Excess of urinal bicarbonate in distal tubule increases excretion of K^+ leading to hypokalemia: Excessive urinal bicarbonates bind Na^+ that is reabsorbed by the principal cells and bicarbonates negative charge stimulate K^+ excretion

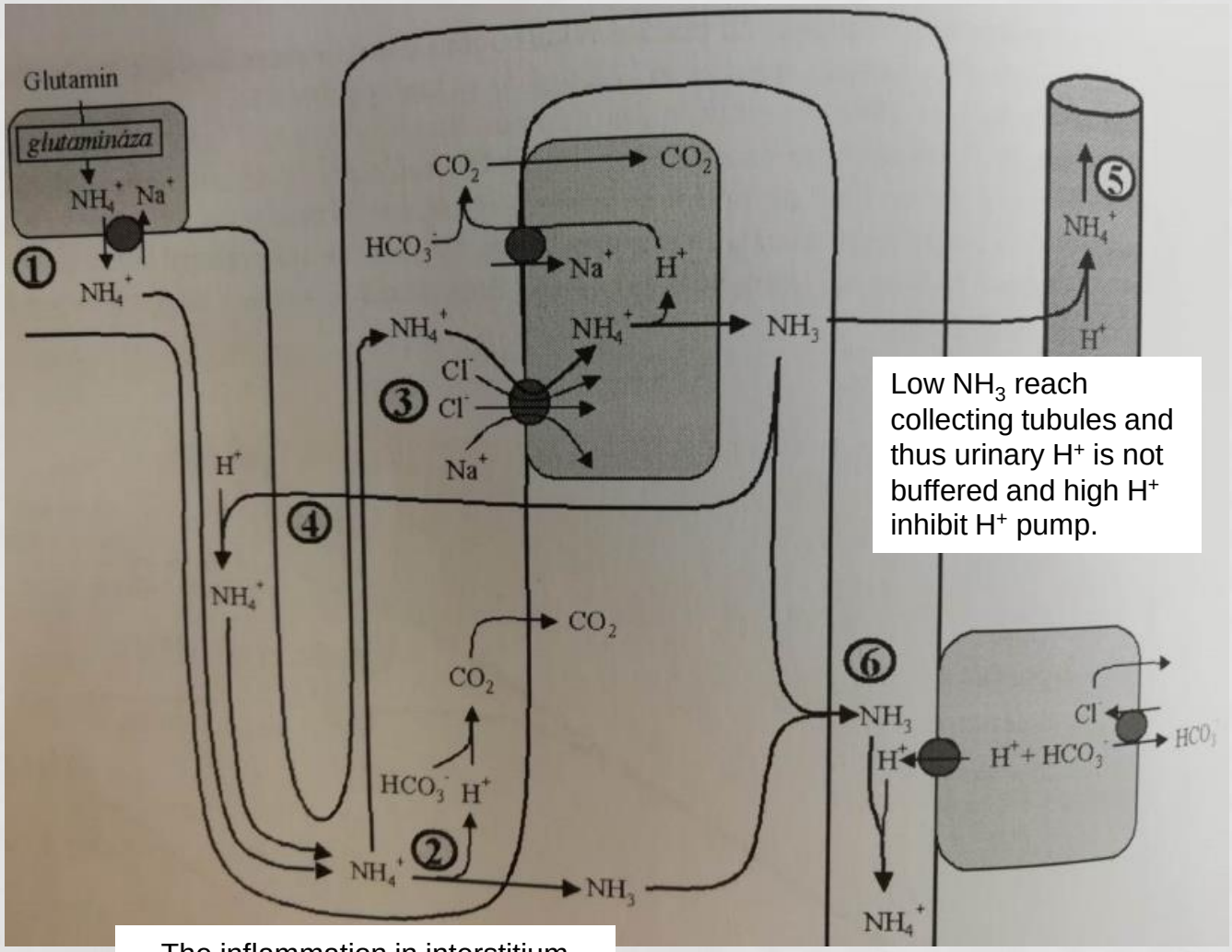
Type III: renal tubular acidosis with decreased glomerular filtration

is caused by low ammonia (NH_3) in distal tubules. The inflammation of renal interstices or low glomerular filtrations leads to ↓ concentration ability in the descendent Henle loop and less concentrated HCO_3^- do not lead to NH_3 formation. Normally NH_3 diffuse to renal interstitium and later collecting tubules where it buffer H^+. In Type III, the increased urinary NH_4^+ is reabsorbed by ascendent Henle loop ($Na^+/2Cl^-/ NH_4^+$) pump to blood (not renal interstices) and contribute to acidosis.

Type III is caused by

- ↓ low glomerular filtration: hyperchloremia, normokalemia with normal AG
- ↓ ↓ low glomerular filtration (uremic end-stage renal failure with hyperphosphatemia, hypocalcemia, secondary hyperparathyroidism and osteodystrophy): normochloremia, hyperkalemia with ↑ AG due to accumulation of anions of strong acids

Type III RTA: the recycling of NH_3



The primary urine is made denser (more urinary HCO_3^-) due to the hyperosmolar interstitium. If primary urine is not concentrated, NH_3 is not produce from NH_4^+ that continues to Henle loop and is more reabsorbed to blood via $Na^+/2Cl^-/NH_4^+$ (K^+) pump.

Type IV: generalized distal tubular acidosis

Problem with Na^+ resorption and K^+ and H^+ excretion due to low aldosterone. It is associated with high amount of salt in urine, hypochloremia and hyperkalemia. AG is normal as hyperchloremia is relative one: ↓ aldosterone decreases Na^+ and Cl^- reabsorption but Na^+ reabsorption is more affected than Cl^- reabsorption.

Type IV is caused by

- Hypoaldosteronism: aldosterone should stimulate H^+ ATPase
- Relative hypoaldosteronism: due to low renin activity in diabetic nephropathy or hypertensive nephrosclerosis

Hypochloremic alkalosis based on Urinary Chloride

Urine chloride < 25 mEq/L

- often associated with volume depletion (contraction alkalosis)
- respond to saline infusion
- common causes:
  - ✓ vomiting gastric content (90% of cases)
  - ✓ alkalotic therapy, such as bicarbonate administrated in peptic ulcer
  - ✓ Diuretic therapy : loop diuretics and thiazide
  - ✓ Cystic fibrosis

Urine chloride > 40 mEq/L

- ✓ often associated with volume expansion and hypokalemia
- ✓ resistant to therapy with saline infusion
- ✓ causes: excess aldosterone, severe K^+ deficiency, current diuretic therapy, Bartter syndrome

VOMITING

hydrochloric acid in the gastric juice

Aldosterone:

- In the principal cells of the distal tubule aldosterone activates the basolateral Na^+/K^+ pumps. Cl^- follow the reabsorbed Na^+ to neutralize e. charge.
- In intercalated cells of collecting tubules aldosterone stimulates secretion of H^+ via the H+/ATPase. It also increases generation of "new" bicarbonate within these same cells, which will be reabsorbed

- acid loss
- water depletion
- depletion of ECF sodium, potassium and chloride.

Low Na^+ induces sodium retention and increases K+ excretion

Low K^+ induces H^+ shift toward intracellular space contributes to alkalosis

Low Cl^-

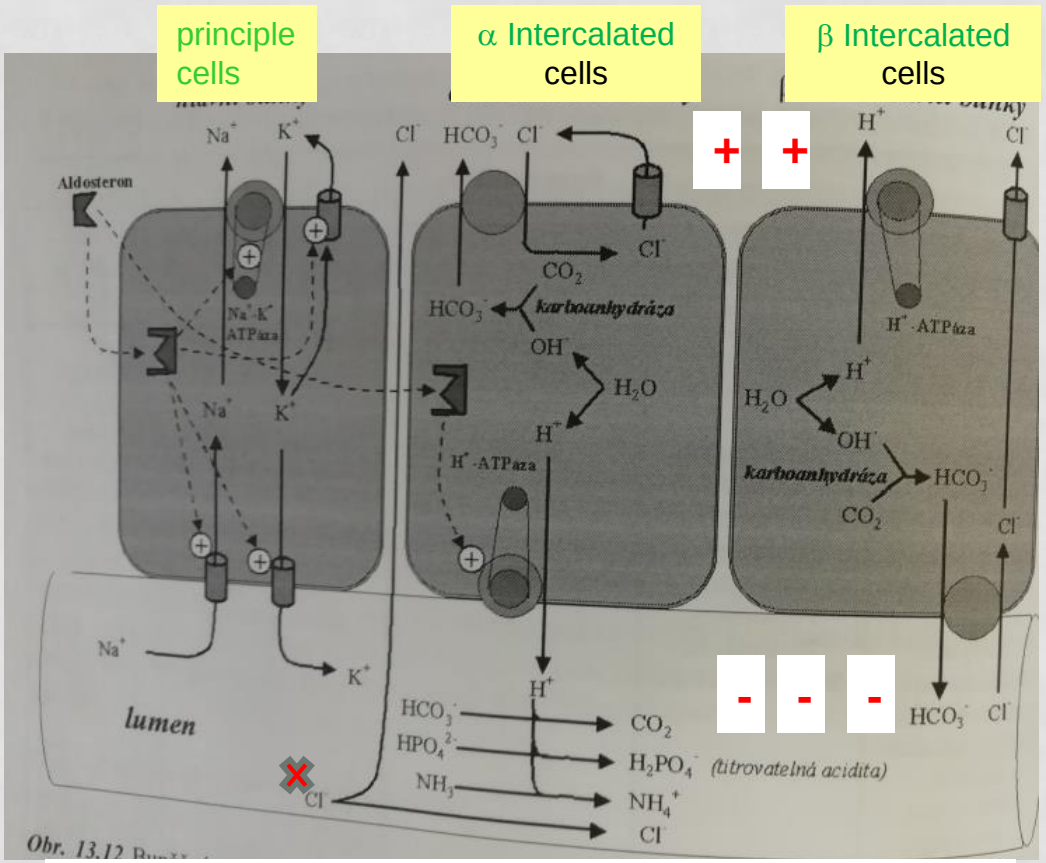
- The kidneys increase HCO_3^- reabsorption to maintain an anionic balance because the ECF Cl^- concentration is decreased. (Like chloride shift: In venous blood, the red cells release Cl^- in exchange for HCO_3^-)
- Low Cl^- cannot neutralize the reabsorbed Na^+ and more positive basolateral side potentiate the effect of aldosterone supporting K^+ and H^+ excretion to urine (hypochloremic alkalosis with paradoxical acidification of urine)

Therapy: Cl^- must be replenisched in infusion t.

Vomiting: the effect of Cl^- loss

Normal situation:
principle cells (aldosterone): Na^+ is absorbed. K^+ is secreted. Passive transport of Cl^- follow Na^+ reabsorption and lumen is less negative but enough negative to stimulate H^+ and K^+ excretion.
Vomiting: ↓ Cl^- potentiate the effect of aldosterone supporting K^+ and H^+ excretion (intercalated cells): hypochloremic alkalosis with paradoxical acidification of urine)

More positive charge without Cl^-



More negative charge support H+ excretion in hypochloremia

Therapy: replenished Cl^- stimulate B-intercalated cells to excrete HCO_3^-

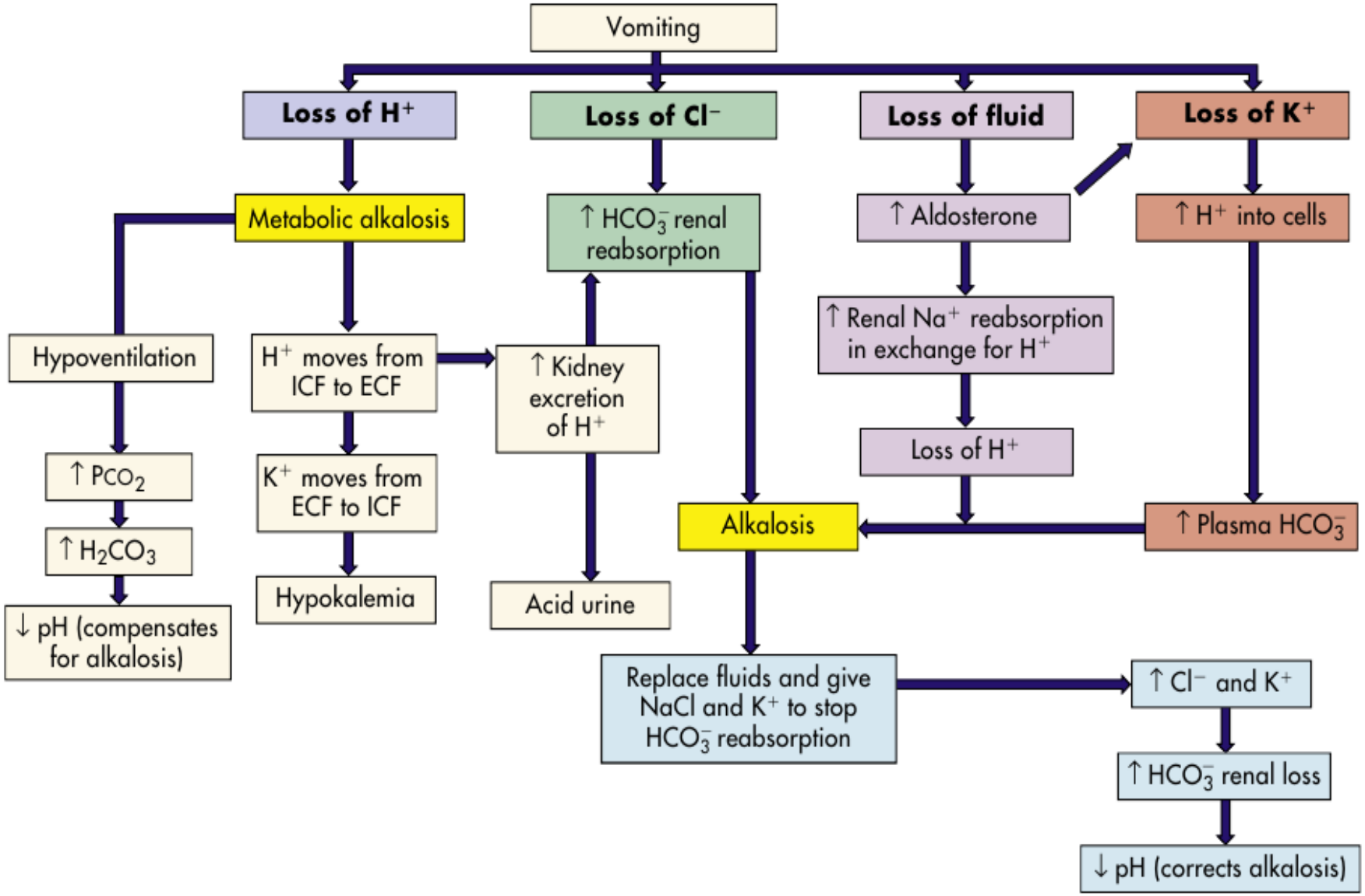
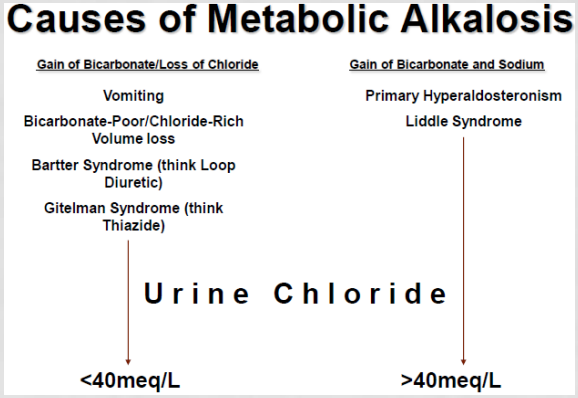


FIGURE 3-17 Hypochloremic Metabolic Alkalosis. See text for abbreviations.

Hyperaldosteronism

primary or secondary (e.g. due to ↓ oncotic pressure in liver cirrhosis) hyperaldosteronism

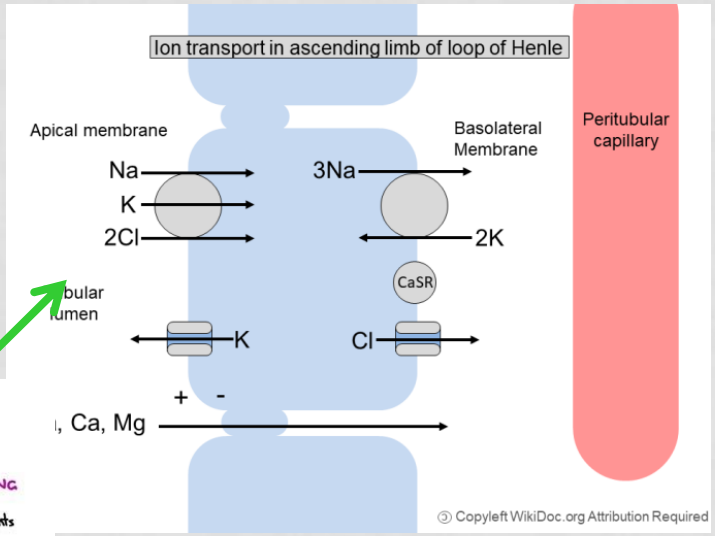
- ↑ aldosterone: ↑ renal excretion of K^+ causing hypokalemia and ↑ H^+ excretion causing m. alkalosis and ↑ Na^+ reabsorption causing ↑ Cl^- reabsorption with low urinary Cl^- (but higher compare that in vomiting).
    - Hypokalemia is also supported by
        - K^+ shift from the extracellular space to intercellular space in exchange for H^+
        - ↑ intracellular K^+ in renal tubular cells that also support K^+ excretion

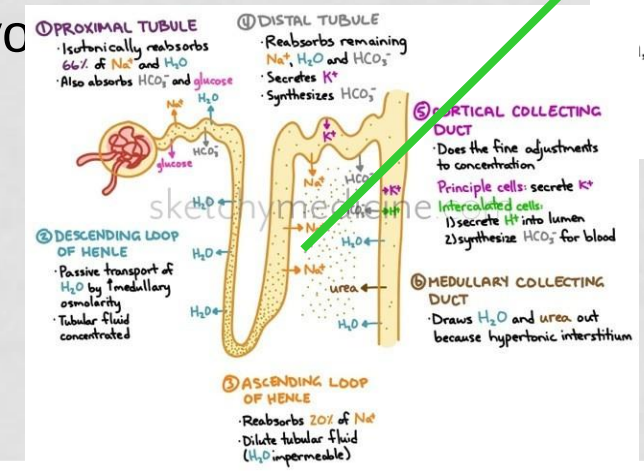


Bartter syndrome (BS)

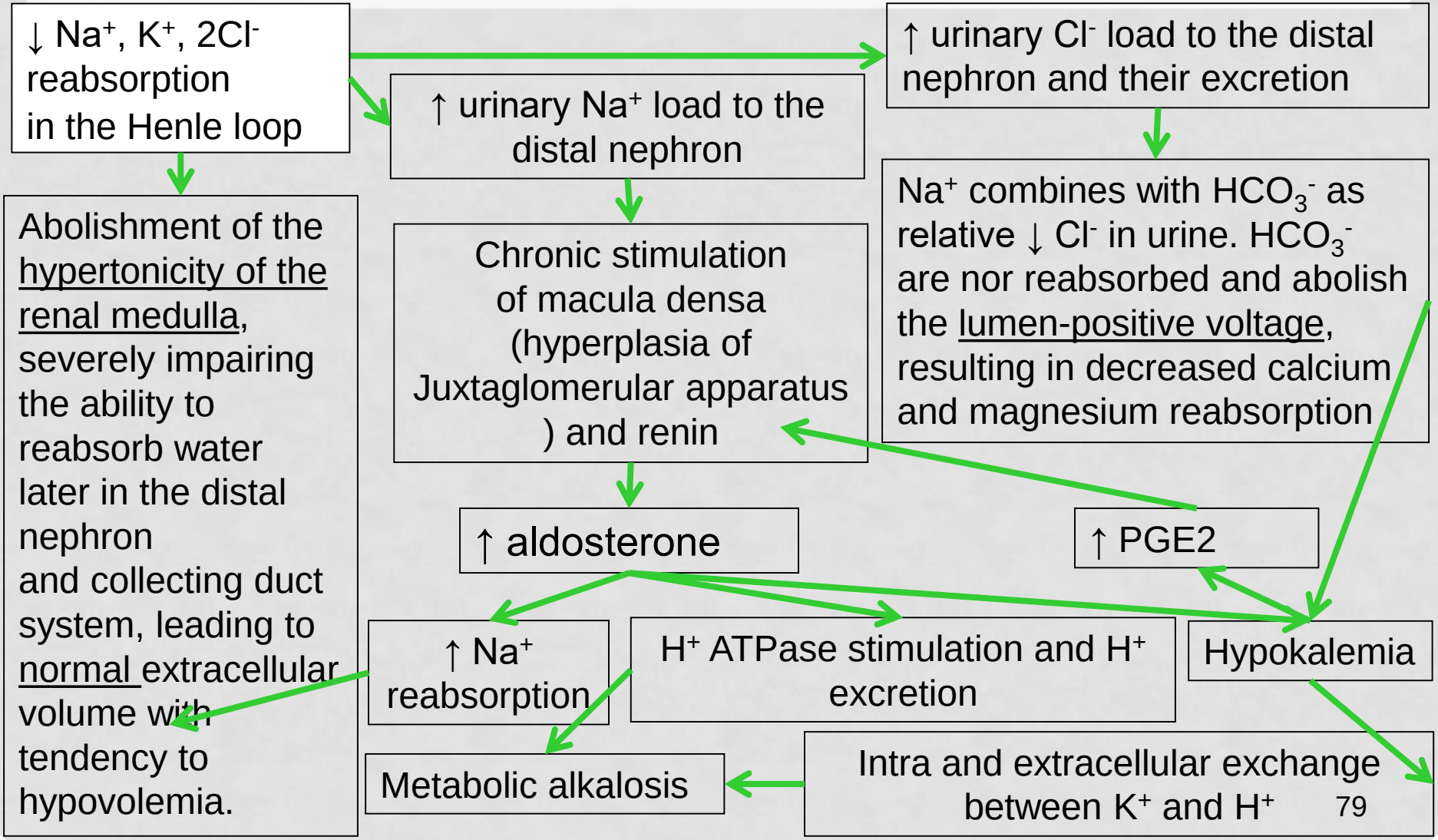
mutations involving the Na^+-K^+-$2Cl^-$ cotransporter in the thick ascending limb of the loop of Henle. Na^+-K^+-$2Cl^-$ cotransport one sodium, one potassium, and two chloride ions across the apical membrane of the tubule.

- Primary hypereninism and secondary hyperaldosteronism
- Hypokalemia
- Hypochloremia
- Metabolic alkalosis
- Tendency to hypovo





Bartter syndrome (BS)



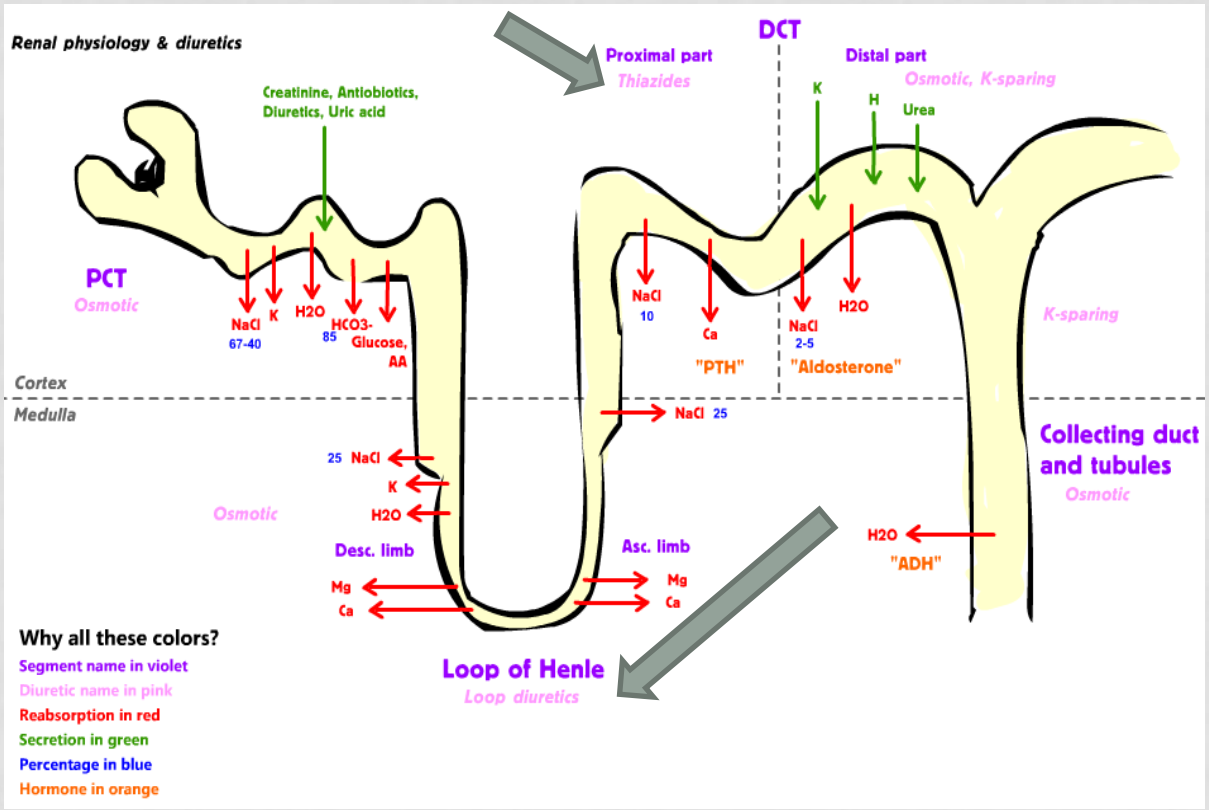
Liddle's syndrome
or **Liddle syndrome**

genetic mutation in epithelial sodium channel that transports sodium into cells. This mutation leads to loss of ability to be degraded and high amounts of the channel being chronically present on the collecting duct epithelium.(hyperaldosteronism-like state)

- high blood pressure
- low plasma renin, normal to low levels of aldosterone,
- kidney: excessive reabsorption of Na^+ and loss of K^+ (metabolic alkalosis, hypokalemia)

Diuretic therapy

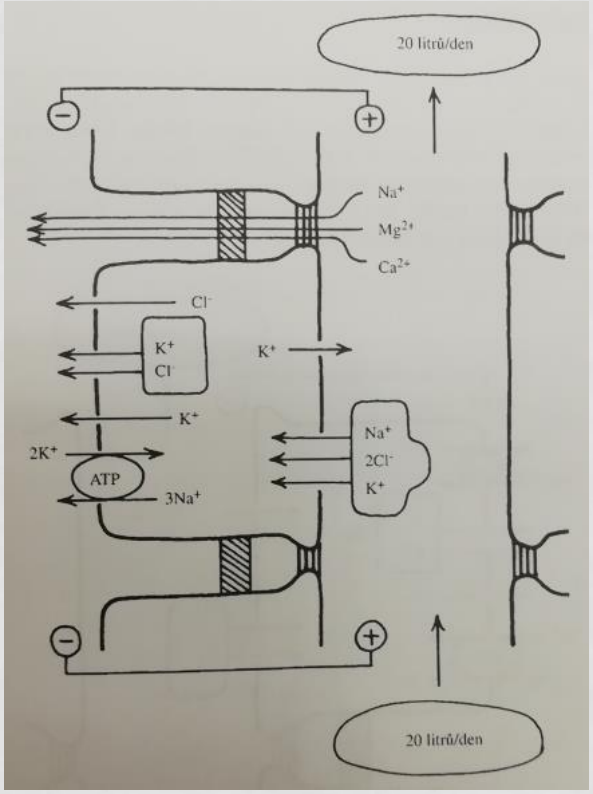
- Loop diuretics: alkalosis (in ascendent loop Na^+-K^+-$2Cl^-$ symporter)
- Thiazide diuretics: alkalosis (in distal tubules Na^+Cl^- symporter)
- (note: inhibitors of carbonic anhydrase induce acidosis)



Diuretic therapy

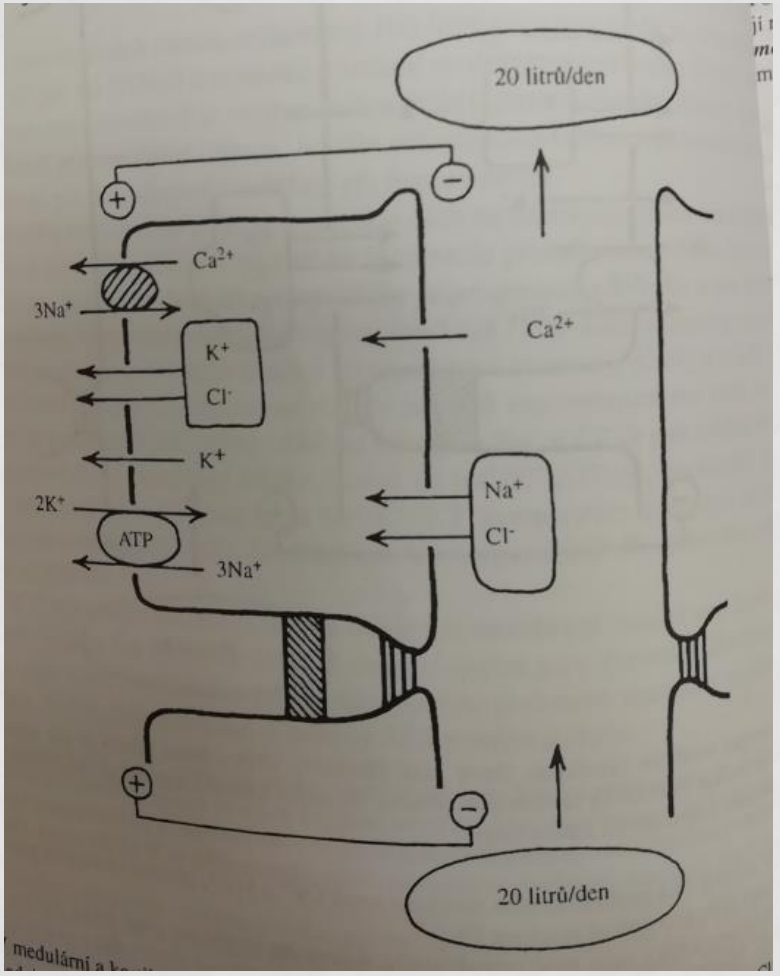
Loop diuretics act on the Na^+-K^+-$2Cl^-$ symporter in the thick ascending limb of the loop of Henle to inhibit sodium, chloride and potassium reabsorption.

- Thus more Na^+ continue to distal tubule where the increased Na^+ increase excretion of $K^+$and H^+ (hypokalemia and alkalosis)
- It also inhibits magnesium and calcium reabsor ption that depends on electric gradient
- Used in treatment of lung edema: decreased venous return to heart



Diuretic therapy

- Thiazide inhibit the sodium-chloride symporter which has the function of reabsorbing sodium and chloride ions causing the body to release NaCl and water
- hypokalemia and alkalosis



Simple (with or without compensation) and Mixed acid base disturbances

Example:

Baseline $HCO3^-$: 24 mEq/L , PCO_2: 40 mmHg, PH: 7.4

- Now: $HCO3^-$: 18 mEq/L, PCO_2: 34 mmHg, PH: 7.34

>> Primary metabolic acidosis with a secondary respiratory alkalosis.

- Now: $HCO3^-$: 18 mEq/L, PCO_2: 30mmHg, PH: 7.4

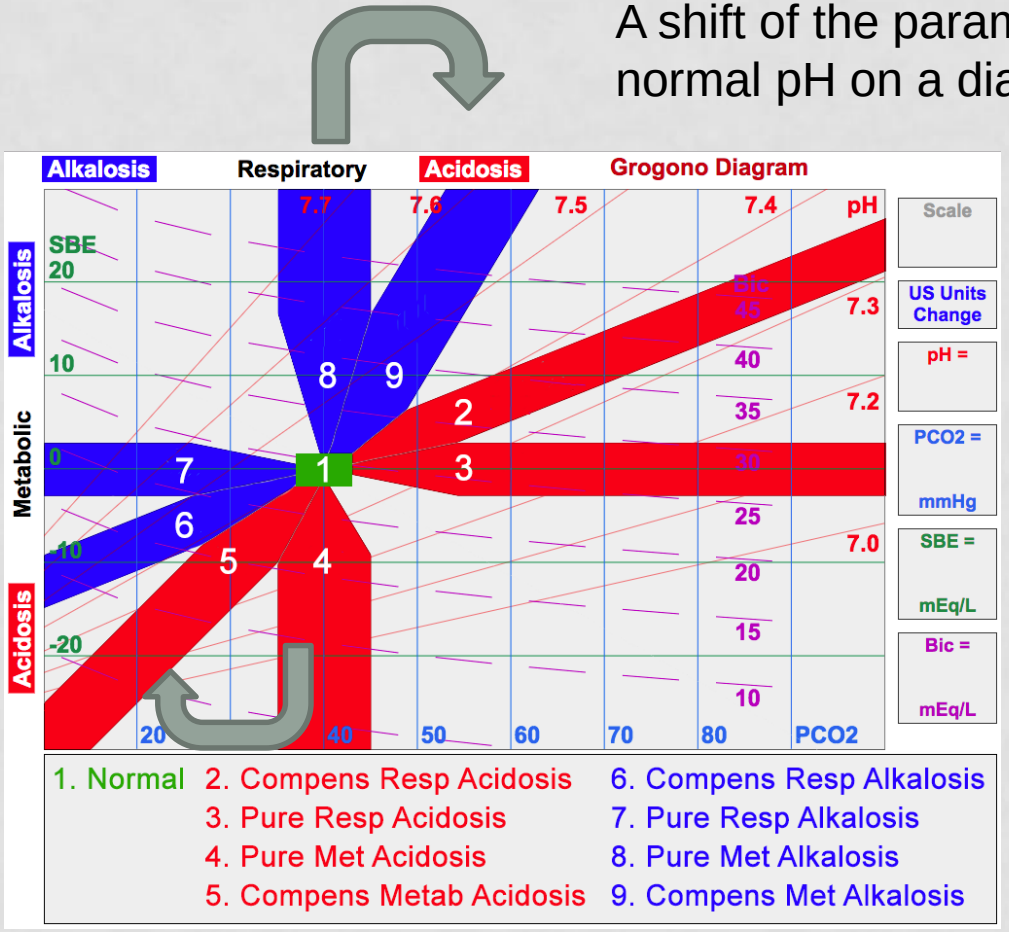
>> Mixed primary metabolic acidosis and primary respiratory alkalosis.

- Now: $HCO3^-$: 18 mEq/L, PCO_2: 37 mmHg, PH: 7.29

>> Mixed primary metabolic acidosis and primary respiratory acidosis.

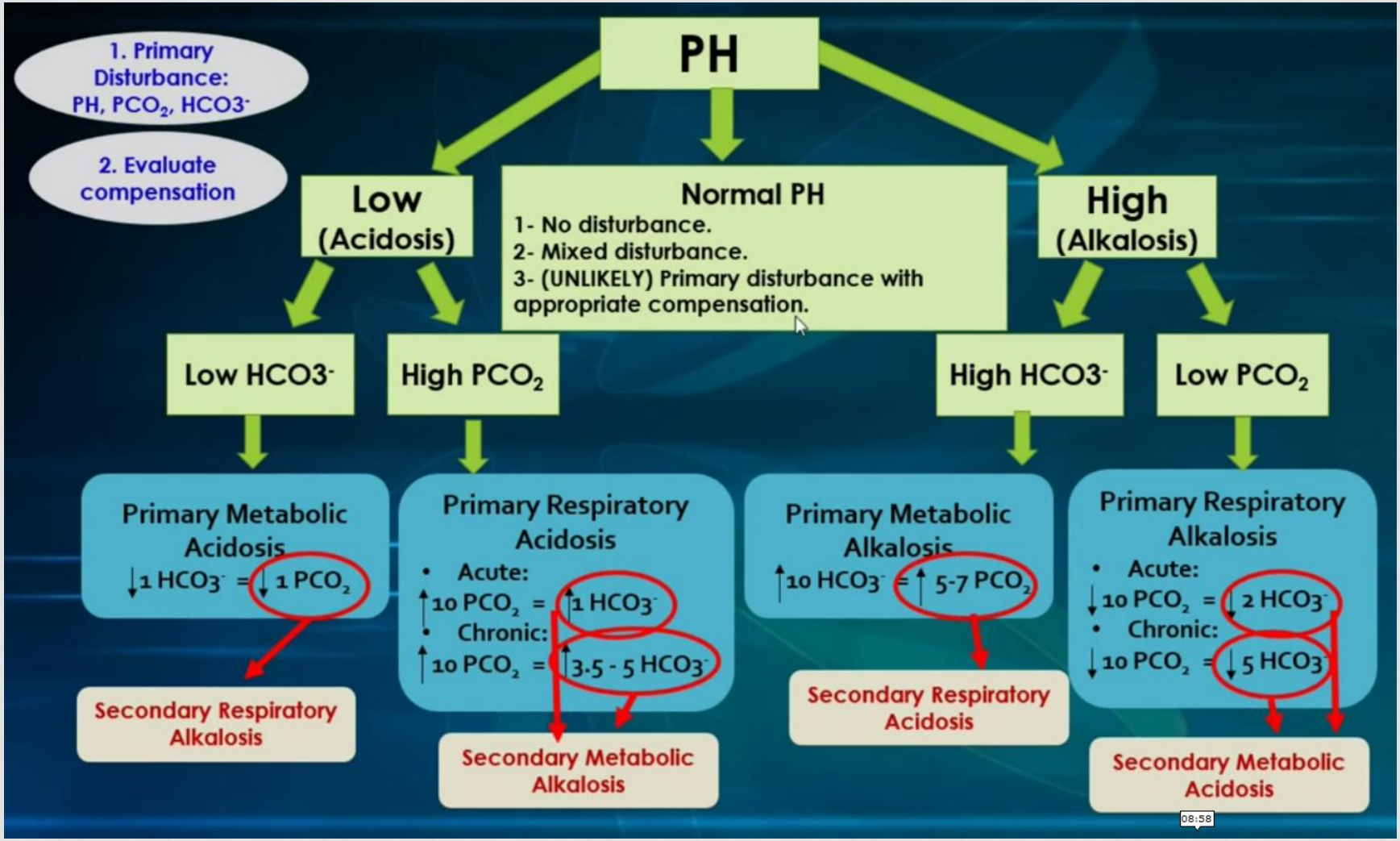
Davenpost diagram

A shift of the parameters toward normal pH on a diagonal



https://www.youtube.com/watch?v=snp5oq-hLpg&t=237s

Normal pH doesn‘t have to be healthy person.



Acidemia: general effect

- Negative inotropic (contractility), and chronotropic (heart rate) effect but it is neutralized by catecholamines that are also stimulated by acidosis.
- Inhibit CNS including respiratory centers
- In systemic vessels,
  - acidosis causes vasodilation in peripheral arterioles but is also offset by catecholamine, so that peripheral vascular resistance remains relatively constant.
    - In brain, vasodilation can cause edema.
  - in the venous system, the direct effect of acidosis is vasoconstriction, which is further enhanced by catecholamine release: prevent lung edema
- Osteoporosis : Ca^{2+} and phosphate liberated from bone (hydroxyapatite, Ca5(PO4)3(OH))
- Final stage: coma (due to H^+-mediated protein denaturation in CNS, dehydration, hyperosmolarity in diabetic ketoacidosis)

Alkalemia: general effect

- Acute alkalosis increases binding and reduces free plasma Ca^{2+} leading to tetany
- Compensation of MA by hypoventilation is limited by O_2
- In brain, vasoconstriction that decrease its perfusion
- Final stage: coma or asphyxia

1. Case

A 65-year-old man comes to the physician with a 3-hour history of shortness of breath after feeling ill for the past week. His BMI is 30 kg/m^2. His temperature is 38.3° C (101° F), pulse is 96/min, respirations are 20/min and shallow, and blood pressure is 145/90 mm Hg.

pH	7.33
PO_2	61 mm Hg
PCO_2	50 mm Hg
HCO_3^-	26 mEq/L

1. Compensations for Respiratory Acidosis

- Acute respiratory acidosis
  - HCO_3 increases by 1 for every 10 mm increases in pCO_2
- Chronic respiratory acidosis
  - HCO_3 increases by 3 for every 10 mm increases in pCO_2

1. Description

R_{acid} acute

pH	7.33
PO_2	61 mm Hg
PCO_2	50 mm Hg
HCO_3^-	**26** mEq/L

1-2 mEq/L increase in HCO_3^- for every 10 mm Hg increase in PCO_2.

PCO_2 increase = 50-40 = 10 mm Hg.

HCO_3^- increase predicted = (1-2) x (10/10) = 1-2 mEq/L

add to 24 mEq/L (reference point) = **25-26** mEq/L

History suggests hypoventilation, supported by increased PCO_2 and lower than anticipated PO_2.

Respiratory acidosis (acute) due to no renal compensation.

Obesity hypoventilation syndrome (OHS) or Pickwickian Syndrome

- is defined as a combination of Obesity (BMI>40 kg/m^2), daytime hypoventilation characterized by hypercapnia and hypoxemia ($PaCO_2$ > 45 mm Hg and PaO_2 < 70 mm Hg at sea level) and sleep-disordered breathing (SDB) in the absence of an alternative cause for hypoventilation like obstructive or restrictive lung disease, chest wall disorders like kyphoscoliosis, neuromuscular disorders and congenital central hypoventilation

2. Case

A 56-year-old woman with COPD is brought to the physician with a 3-hour history of shortness of breath. Her temperature is 37° C (98.6° F), pulse is 90/min, respirations are 22/min and shallow, and blood pressure is 135/80 mm Hg.

pH	7.39
PO_2	62 mm Hg
PCO_2	52 mm Hg
HCO_3^-	29 mEq/L

2. Compensations for Respiratory Acidosis

- Acute respiratory acidosis
  - HCO_3 increases by 1 for every 10 mm increases in pCO_2
- Chronic respiratory acidosis
  - HCO_3 increases by 3 for every 10 mm increases in pCO_2

2. Description

R_{acid} chronic

pH	7.39
PO_2	62 mm Hg
PCO_2	52 mm Hg
HCO_3^-	**29** mEq/L

3-4 mEq/L increase in HCO_3^- for every 10 mm Hg increase in PCO_2.

PCO_2 increase = 52-40 = 12 mm Hg.

HCO_3^- increase predicted = (3-4) x (12/10) = 4-5 mEq/L

add to 24 mEq/L (reference point) = **28-29** mEq/L

History suggests hypoventilation, supported by increased PCO_2.
Respiratory acidosis (chronic) with renal compensation.

3. Case

A 17-year-old woman is brought to the physician with a 3-hour history of epigastric pain and nausea. She admits taking a large dose of aspirin. Her respirations are 20/min and full.

pH	7.55
PO_2	104 mm Hg
PCO_2	25 mm Hg
HCO_3^-	22 mEq/L

3. Compensations for Respiratory Alkalosis

- Acute respiratory alkalosis
  - HCO_3 decreases by 1-2 for every 10 mm decrease in pCO_2
- Chronic respiratory alkalosis
  - HCO_3 decreases by 4 for every 10 mm decrease in pCO_2

3. Description

R_{alk} acute

pH	7.55
PO_2	104 mm Hg
PCO_2	25 mm Hg
HCO_3^-	**22** mEq/L

1-2 mEq/L decrease in HCO_3^- for every 10 mm Hg decrease in PCO_2.

PCO_2 decrease = 40-25 = 15 mm Hg.

HCO_3^- decrease predicted = (1-2) x (15/10) = 2-3 mEq/L

subtract from 24 mEq/L (reference point) = **21-22** mEq/L

History suggests hyperventilation, supported by decreased PCO_2.

Respiratory alkalosis (acute) due to no renal compensation.

4. Case

A 81-year-old woman with a history of anxiety is brought to the physician with a 2-day history of shortness of breath. She has been living at 9,000 ft elevation for the past 1 month. Her respirations are full at 20/min.

pH	7.44
PO_2	69 mm Hg
PCO_2	24 mm Hg
HCO_3^-	16 mEq/L

4. Compensations for Respiratory Alkalosis

- Acute respiratory alkalosis
  - HCO_3 decreases by 1-2 for every 10 mm decrease in pCO_2
- Chronic respiratory alkalosis
  - HCO_3 decreases by 4 for every 10 mm decrease in pCO_2

4. Description

R_{alk} chronic

pH	7.44
PO_2	69 mm Hg
PCO_2	24 mm Hg
HCO_3^-	**16** mEq/L

4-5 mEq/L decrease in HCO_3^- for every 10 mm Hg decrease in PCO_2.

PCO_2 decrease = 40-24 = 16 mm Hg.

HCO_3^- decrease predicted = (4-5) x (16/10) = 6-8 mEq/L

subtract from 24 mEq/L (reference point) = **16-18** mEq/L

History suggests hyperventilation, supported by decreased PCO_2.

Respiratory alkalosis (chronic) with renal compensation.

5. Case

A 75-year-old man with severe congestive heart failure is brought to the emergency department. He takes none of his prescribed medications. His respirations are 24/min and blood pressure is 80/50 mm Hg. He has decreased urine output; his baseline creatinine concentration has been 1.6 mg/dL.

pH	7.19	Na^+	135 mEq/L
PO_2	80 mm Hg	K^+	4.0 mEq/L
PCO_2	21 mm Hg	Cl^-	97 mEq/L
HCO_3^-	8 mEq/L	CO_2, total	8 mEq/L
Lactate	20 mEq/L	Urea	54 mg/dL
		Creatinine	2.5 mg/dL

5. Compensations for Metabolic Acidosis

- Metabolic acidosis with appropriate respiratory compensation
- 1.2 mm Hg decrease in PCO_2 for every 1 mEq/L decrease in HCO_3^-.

M_{acid} increased AG

5. Description

pH	7.19	Na^+	135 mEq/L
PO_2	80 mm Hg	K^+	4.0 mEq/L
PCO_2	**21** mm Hg	Cl^-	97 mEq/L
HCO_3^-	8 mEq/L	CO_2, total	8 mEq/L
Lactate	20 mEq/L	Urea	54 mg/dL
AG = 135-97-8=30 mEq/L		Creatinine	2.5 mg/dL
DR = (30-12)/(24-8) = 1.1			

1.2 mm Hg decrease in PCO_2 for every 1 mEq/L decrease in HCO_3^-.

HCO_3^- decrease = 24-8 = 16 mEq/L

PCO_2 decrease predicted = 1.2 x 16 = 19 mm Hg.

subtract from 40 mm Hg (reference point) = **21** mm Hg

History suggests congestive heart failure (poor perfusion).

Metabolic acidosis with appropriate respiratory compensation.

Strong Ion Difference (SID)

- Another approach to assess acid-base disturbance that is based on the difference between the concentrations of strong cations and strong anions. (Astrup measures pH, HCO_3^- and pCO_2)
  - SID = [strong cations] – [strong anions]
  - apparent SID = SIDa = $(Na^+ + K^+ + Ca^{2+} + Mg^{2+}) – (Cl^– + \text{L-lactate} + \text{urate})$

- In normal human plasma the SID is 42 mEq/L
- the number of positive and negative ions in a solution must be equal (SID = 0), so there are unmeasured anions
  - increased SID (>0) leads to alkalosis (increase in unmeasured anions)
  - decreased SID (<0) acidosis

References

Look up the particular theme in the following indexes

- Porth's Pathophysiology: Concepts of Altered Health States (9th Edition) by Sheila Grossman, Carol Mattson Porth. Wolters Kluwer Health | Lippincott Williams & Wilkins, 2014.
- McCance K. L., Huether S. E.: Pathophysiology. 8th Edition. Mosby, 2018.
- Silbernagl S, Lang F. Color Atlas of Pathophysiology, 3rd Ed. Thieme, 2016.
- https://pfyziol.upol.cz/ - selected text passages (according to teachers' recommendations)